Large Language Models in Islamic Jurisprudence: A Multi-Method Framework for Hadith Authentication and Hanafi Fatwa Synthesis

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Statement of Originality

I hereby declare that this dissertation is my own work and that, to the best of my knowledge and belief, it contains no material previously published or written by another person, except where due acknowledgment has been made in the text.

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Abstract

The rapid paradigm shift from traditional Machine Learning toward Generative Large Language Models (LLMs) has profoundly disrupted educational technology. However, deploying Generative AI in highly specialized, authoritative domains—such as Islamic Jurisprudence (Fatawa) and classical Hadith sciences—presents severe risks, as model hallucination constitutes an unacceptable doctrinal failure. This dissertation investigates the complete evolutionary trajectory of AI in specialized education, beginning with baseline Natural Language Processing (NLP) applications for text extraction and a Systematic Literature Review mapping pre-generative educational AI. Recognizing that base LLMs cannot natively satisfy the zero-hallucination requirement of Islamic theology (proven via an empirical LLM-as-a-judge experiment yielding $< 50\%$ accuracy on Hanafi jurisprudence), this research engineers a systemic solution: the "Agentix Islam RAG" framework. By creating entirely new structured datasets (the Ahadith Authenticity Dataset and the Hanafi Fatwa Dataset), establishing strict Table of Contents (TOC) hierarchy embeddings, and utilizing context-compressed Retrieval-Augmented Generation (RAG), the system effectively forces generative models to reason strictly within authoritative classical texts. The deployed agentic architecture successfully answered over 4,000 real-world theological queries with complete doctrinal accuracy, demonstrating that strict computational guardrails can successfully align advanced AI with zero-tolerance religious domains.

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List of Abbreviations

AI	Artificial Intelligence
LLM	Large Language Model
NLP	Natural Language Processing
RAG	Retrieval-Augmented Generation
SLR	Systematic Literature Review
API	Application Programming Interface
TOC	Table of Contents
JSON	JavaScript Object Notation

Chapter 1

Introduction

1.1 Relevance of the Topic

The advent of Large Language Models (LLMs) has fundamentally disrupted educational technology. While traditional Natural Language Processing (NLP) historically augmented education through static models (e.g., grading prediction systems, which formed the early phases of this research), the generative capability of modern AI presents both unprecedented opportunities and severe risks. In specialized, authoritative domains—such as Islamic Jurisprudence (Fatawa) and classical Islamic literature—model hallucination is not merely an error; it represents a critical failure of doctrinal integrity. Thus, developing reliable, bounded ”Agentic” architectures capable of querying classical texts without hallucination is exceptionally relevant to modern theological education.

1.2 Degree of Development in the Subject Area

A baseline derived from the Systematic Literature Review in **R2**, charting the state of AI in education prior to the generative AI explosion, and identifying the specific gap in addressing highly sensitive, low-resource classical Arabic frameworks.

1.3 Purpose and Objectives of the Research

The primary purpose of this research is to architect and validate a highly reliable, LLM-based agentic system capable of answering complex Islamic inquiries by directly reasoning over classical texts.

- **Objective 1:** To establish baseline capabilities in processing and classifying canonical Islamic texts (Ahadith AI - **R3**).
- **Objective 2:** To conceptualize and integrate AI natively into the Islamic Fatawa resolution process (Fatawa AI - **R5**).
- **Objective 3:** To deploy a robust Retrieval-Augmented Generation (RAG) web application that enforces zero-hallucination policies through structural database engineering (Agentix Islam RAG - **R6**).

1.4 Scientific Novelty

The scientific novelty lies in designing a multi-agent framework constrained by strict theological guardrails, successfully mitigating LLM hallucination in a zero-tolerance domain. This relies heavily on novel methodologies for structuring complex, un-digitized classical Arabic texts into hierarchical, semantically searchable geometries—specifically, the creation of the **Ahadith Authenticity Dataset (R3)** and the structured computational **Hanafi Fatwa Dataset (R4/R5)**. Additionally, the development of Agentix Islam (**R6**), a RAG system that can embed huge classical books, creates an Islamic Vectorized Database, and answers questions about Islam using contextual RAG and LLMs.

1.5 Theoretical and Practical Significance

- **Theoretical:** Extends the literature on Agentic RAG systems by demonstrating effective truncation of generative hallucinations in non-Western, highly sensitive domains.
- **Practical (Dataset Contributions):** The creation and structuring of vast, high-quality Islamic datasets—namely the Ahadith classification dataset (**R3**) and the comprehensive Hanafi Fatwa dataset (**R5**)—parsed from classical books, converted into hierarchical JSON objects, and embedded into Vector databases for future researchers. The deployment of a functional Web Application directly aiding scholars and students.

Chapter 2

Traditional AI Applications in Education: NLP and Transitional Words

2.1 Introduction

In the present day, accessibility and power of artificial intelligence grow with every day. AI is applied in a variety of fields. If, for instance, we consider software engineering, AI tools can be used at almost every step of the process: requirement elicitation [1], development [2], testing [3] and customer service [4].

Of these applications, ones that stand out are these which require to process text - and not the uniform type of text like code in a particular programming language or sanitized data set, but natural language, such as elicited requirements, text-based feedback or text from a scientific paper.

For these cases, a technology known as natural language processing (NLP) is often used [5]. NLP is a subfield of AI that focuses on enabling machines to understand, interpret, and generate human language. Its main application is analyzing and processing natural language data to extract certain information - for instance, patterns or tokens (i.e.

locations, names or numbers).

In the following chapters, we will discuss existing applications of NLP in language processing and provide a solution to one particular issue - detecting transitional words in a given scientific text.

2.2 Literature review

2.2.1 NLP applications in software engineering

When it comes to software engineering, NLP is mostly applied to documentation, requirements and feedback, as these are the artifacts that are usually represented in the form of a natural text.

Di Sorbo, Visaggio, Di Penta, Canfora and Panichella [6] identified that processing unstructured artifacts is useful for development of suggester systems (which, in turn, help in several different ways, such as extracting action points from user feedback and generating documentation). However, such processing requires a manually-specified ruleset. To mitigate this, researchers used NLP and created a system that can produce rules automatically, with more than a third of rules generated by their system being relevant for a specified task.

Al Omran and Treude [7] discuss application of NLP to software documentation in multiple forms (such as StackOverflow data and GitHub ReadMe files) and conclude that this application of NLP can lead to a discovery of useful and actionable information, but requires justification (as certain actions could be performed more efficiently by approaches such as text mining) as well as informed choice (and, potentially, further customization) of an NLP library.

Ferrari and Esuli [8] discuss how the ambiguity during the requirement elicitation process can negatively affect later stages of development and propose an NLP-based solution to this problem. A model was created for each area of stakeholders' expertise and later used to automatically detect and score ambiguity of terms with varying meanings in different areas. This approach was found to be applicable to most of the scenarios in which it was tested.

Joshi and Deshpande [9] proposed that current tools for automatic generation of UML diagrams from natural language requirements are inefficient and developed an NLP-based tool called RAUE that implements natural language processing for extracting data for UML diagram generation. This tool was able to successfully detect all types of relations between entities with an accuracy of about 85% (based on precision and recall metrics).

2.2.2 NLP applications in text analysis

When it comes to text analysis, NLP is used in two major ways: either for sentiment analysis (mostly as one step in the process - i.e. tokenization) or for extractive analysis.

Sentiment analysis

Sentiment analysis is mostly concerned with what is the general meaning of a particular text. These approaches try to abstract from the text in general towards the intent of the author. In these cases, NLP is mostly used as a tokenizer or in other data cleanup approaches.

Rameshbhai and Paulose [10] discuss one such application of NLP - newspaper headlines opinion mining. In this paper, NLP is used for tokenization of data to later on be processed manually and later produce a model using SVM.

Hasan, Maliha and Arifuzzaman [11] also use NLP in sentiment analysis. Here, NLP is used to clean data as well as to detect and remove stop-words. After sanitizing data in this way, other techniques, such as Bag of Words (BoW) and Term Frequency-

Inverse Document Frequency (TF-IDF) are used for the sentiment analysis itself.

Extractive analysis

Extractive analysis is, as name would suggest, largely focused on extracting data from provided text and compiling it in a certain way. Here, applications of NLP are of much greater variety, as a lot of data can be extracted from text using NLP (part-of-speech data, lemmas, tokens, etc.).

Hendrycks, Burns, Chen and Ball [12] concluded that there are a lack of domain-specific datasets, and created Contract Understanding Atticus Dataset (CUAD) - a dataset of other 13000 manually written annotations, which is created with the purpose of highlighting important parts of legal contracts for a human to review. In addition to its main purpose, CUAD is usable as a benchmark for domain-specific NLP models.

Carchiolo, Longheu, Reitano and Zagarella [13] applied NLP to process scanned medical prescriptions. Their implementation does so by extracting embedded terms and then comparing them against patient and doctor personal data, symptoms, diagnosis and proposed treatment. Resulting system was able to automatically classify 95% of prescriptions as either grantable or not.

Sarica, Song, Low and Luo [14] proposed an approach to improve grant search. They use NLP to train a model, which is later used to rank keywords in the patent information. This way, the system can provide a list of a potentially relevant terms to include in the future searches, which allows users to greatly improve their searching progress. Moreover, this system also provides groups of keywords that are relevant to the topic only in combination with each other, but not by themselves.

Lastly, Kauchak, Leroy, Pei and Colina [15] discuss a topic that is the closest in scope to that of this paper: predicting transitional words between pairs of sentences. They considered two questions for a given pair of sentences: whether a transition should be used at all and, if it is, which one would fit the context the most. Work was performed for English and Spanish languages, and average accuracy of transitional word choice were 78% and 84%, respectively.

2.2.3 Takeaways

Based on the existing work research, following conclusions can be drawn:

1. NLP has been used in a variety of text-analysis scenarios to a relatively high success
2. NLP shows best results when applied to problems which require detection of words relevant to a specific topic or category
3. While relatively accurate, NLP-based solutions have accuracy that is in the range of 75-85%, becoming lower in the particularly niche cases. That means, that in simpler projects other approaches might be preferable for greater accuracy, but NLP offers one of the greatest flexibility when it comes to data.

This work is also relevant, because while each of the points stated below has been addressed at least once, there are no research which covers all of them at once:

1. Using NLP for extractive analysis
2. Applying extractive analysis to research papers
3. Detecting a particular non-standard (not covered by pre-trained models) token (transitional word, in this case)

2.3 Implementation

To train the model, following steps had to be taken:

- Decide on the implementation: which libraries have to be used, which models are we planning to use and produce as the result of this project.
- Process transition words data. In this step, we take initial transitions set in a CSV format, process and clean it, and then handle complex transition cases (i.e. transitions consisting on multiple words)
- Generate training data based on sanitized sample text and processed transition words. In this step, we update existing spaCy model with new entities - transitions extracted in the previous step, and use this model to process sanitized data and produce training data (so-called gold standard)
- Train the model using generated training data. In this step, we train the model using spaCy's training module and the gold standard produced in the previous step. After this step, our result model is ready to be used
- Run the model. In this step, we process data previously unused in the training process to test our model
- Assess produced results and relevant metrics. In this step, we compare produced result to the gold standard and use the result to generate confusion matrix, as well as model metrics (accuracy, precision, recall and F1)

2.3.1 General project information

In the implementation of the project, following libraries were utilized:

- spaCy is the NLP library of choice for this project. It is used for the majority of the steps listed above. It was chosen over NLTK due to several advantages:
  - SpaCy has faster text processing speed compared to NLTK
  - NLTK is string based, as opposed to object-based. Processing tokens as

objects simplifies the training process and visualization (as all of the data relevant to the token is stored in a simple to access way)

- displacy is the module of spacy that is used to output formatted data. It is mostly used for visualization
- random is a standard python library. It is used to shuffle training data during the process of model training
- json library is used to save and load intermediary data

During the course of the project two NLP models are used:

- en_core_web_md is the default medium-size english model provided with spaCy. A medium size model is used to strike a balance between speed of use and robustness
- transition_ner is the intermediary model based on the default spaCy model with added entities for labels
- transitions_ner_model is the model trained during the course of the project

2.3.2 Process transition words data

The first step was to process transition words provided to use later on. Originally words were provided in the csv (comma-separated values) format:

```
To show an alternative,alternatively,otherwise,or,if,unless,,,,,,,,,,,,,,,,,,,,,,,,,,,,,
To identify or clarify,that is,in other words,specifically,namely,i.e.,,,,,,,,,,,,,,,,,,,,,,,,,,,,,
To reinforce,in fact,indeed,of course,clearly,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,
```

Figure 2.1: Raw data in CSV format

To process this file, several steps were taken, which could be roughly separated into 3 blocks:

- Reading and cleaning file

- Processing elements with word variations
- Processing elements with word omissions

Implementations of listed steps are shown in the image above.

```python
words = []
complex_cases = []

with open ("files/transition_words.csv", "r") as f:
    text = f.read()
    for s in text.split('\n'):
        words += s.split(',')[1:]

words = [x for x in words if x != ''] # Pass to remove empty entries

for i, w in enumerate(words):
    if ('/' in w): # Expanding elements with multiple variations (i.e. 'more/most importantly')
        sp = w.split(' ')
        ti = [idx for idx, s in enumerate(sp) if '/' in s][0] # Find index of element with slash
        vars = sp[ti].split('/') # Get all variations separated by slash
        for i in range(len(sp)):
            if i != ti:
                for j in range(len(vars)):
                    vars[j] = sp[i] + ' ' + vars[j] if i < ti else vars[j] + ' ' + sp[i]
        words += vars
    if ('...' in w): # Handling strings with ...
        if w[-3:] == '...':
            words[i] = w[:-3].lower()
        else: # Complex case - ... denotes arbitrary amount of text in the middle
            complex_cases.append(w.lower().split('... '))


words = [x for x in words if '/' not in x and '...' not in x] # Removing previously split elements
words += [x.capitalize() for x in words] # Adding capitalized versions (for sentence starts)
```

Figure 2.2: Code for processing data

In the first block, we open the file, read the data from it, and then split values on line feeds (

n) and commas. When we split the file based on commas, empty values are generated (as seen on screenshot of initial data, there are multiple trailing commas, which contain empty values after the split), so we remove those in the last step. We also remove the first entry in each row, as it contains the name of the transition word category, and is not used.

By the end of the first block, certain values are already processed and ready to be used. Those are single word (i.e. similarly, equally) or multiple word (i.e. as a consequence) values In the second block, we process one of the more complex cases - values with multiple word variations (i.e. more/most importantly). To do that, we first find the index of an element with multiple variations (via it containing a slash character). After that, we split this element to obtain all of the word variations. As a final step, we go through the original value, substitute the element containing multiple variations with a particular variation and add it to the result array. This is performed for each of the variations.

In the next block another complex case is processed - elements with omitted words (i.e. It is clear that...). There are two possible cases of such elements: they either contain "..." at the end of the sentence (in which case it is simply trimmed from the element) or they contain "..." in the middle of the sentence (in that case, an array of two elements - part before and part after the dots, - is added to the complex_cases array. It's not currently used, but might be implemented in the future).

After these blocks, two more operations are performed on the array: all of the previously processed complex cases are removed and capitalized versions of every entry is added (to detect entries at the start of the sentence)

2.3.3 Generating training data

In the next step, we use prepared transition words to update existing spaCy model with new entities and use this model to process sanitized data, producing training data used in the next steps as a result

Code for this step can also be separated into 3 blocks:

- Update existing model with new entities
- Load the text
- Process loaded text using the updated model

In the first block, we specify the name of a new label. Then we remove and re-add the necessary processing step (called "pipe" in spacy) to ensure that it is present, but not duplicated in the pipeline. After that we add all of our transition words in a special format used by spaCy (a Python dictionary with keys for label - category of added word, and pattern - specific word or regular expression to match).

In the second block, we load the text and split it into sentences using spaCy

```python
LABEL = "Transition"
if nlp.has_pipe("entity_ruler"):
  nlp.remove_pipe("entity_ruler")
ruler = nlp.add_pipe('entity_ruler', before="ner")
pattern = []
# Adding Labels
for text in words:
  pattern.append({"label": LABEL, "pattern": text})
ruler.add_patterns(pattern)
data_set = []
text = ''

with open ("files/data-prep/abstracts.txt", "r") as f:
  text = f.read()

doc = nlp(text)
sents = [sent.text for sent in doc.sents]

for sent in sents:
  doc = nlp(sent)
  results = []
  entities = []
  for ent in doc.ents:
    if (ent.label_ == LABEL):
      entities.append((ent.start_char, ent.end_char, ent.label_))
  if len([e for e in doc.ents if e.label_ == LABEL]) > 0:
    data_set.append([sent, {"entities": entities}])

with open("files/transition_training_data.json", "w", encoding="utf-8") as f:
  json.dump(data_set, f, indent=4)
```

Figure 2.3: Code for generating training data

In the third step, we process each sentence by finding all of the entities contained in it (both entities from the default model and our transition entities). We then record start position, end position and label of each of the transitions (this is the format of the annotation used in the data set later on). Then we append a list consisting of the sentence and dictionary containing annotations for all of the transition entities found in this sentence (if none were found, the sentence is omitted)

Finally, we export the resulting data in the JSON format to be used later

2.3.4 Training the model

In this step, all of the data is prepared and the model is ready to be trained. This step consists of 2 blocks of code:

- Create and prepare empty model
- Train the model

In the first block we load our previously prepared training data. After that, we create a new empty model. The only pipe used in it is ner (named entity recognition). We add labels to be recognised (in this particular case it's only the "Transition" label, but this

```
ITER=30

with open("files/transition_training_data.json", "r", encoding="utf-8") as f:
    train_data = json.load(f)

nlp = spacy.blank('en')
if "ner" not in nlp.pipe_names:
    ner = nlp.add_pipe("ner")
for _, annotations in train_data:
    for ent in annotations.get("entities"):
        ner.add_label(ent[2])
other_pipes = [pipe for pipe in nlp.pipe_names if pipe != "ner"]
with nlp.disable_pipes(*other_pipes):
    optimizer = nlp.begin_training()
    for itn in range(ITER):
        print("Starting iteration " + str(itn))
        random.shuffle(train_data)
        losses = {}
        examples = []
        for text, annotations in train_data:
            examples.append(spacy.training.Example.from_dict(nlp.make_doc(text), annotations))
        nlp.update(
            examples,
            drop=0.2,
            sgd=optimizer,
            losses=losses
            )
nlp.to_disk("transitions_ner_model")
```

Figure 2.4: Code for model training

code can accommodate additional labels if needed).

At the start of the second block, we disable all of the other pipes except for “ner”, and then start the training process. The training process basically consists of using the model to process a sentence in the text form and then compare it to the gold standard annotations. Then, based on the examples produced that way (using spaCy’s training.Example.from_dict method), the model is updated and one training iteration is finished. In total, this training session took 30 iterations.

After the model is trained, it is saved to the disk. From this moment, it can be imported and used just as default spaCy models

2.3.5 Running the model

By this point, we can run the model on a text that was not used in the previous training (and thus, completely unknown by the model) and successfully detect transition words in the sentences. Example of processed data (formatted using displacy library) is shown below. Each of the highlight words was classified as a transition by our newly trained model.

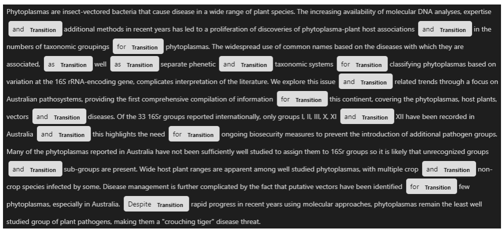

Figure 2.5: Model output visualized via displacy

2.3.6 Assessing produced results and relevant metrics

To gauge performance of our model versus pre-trained spaCy models, we would need to collect data and calculate certain metrics. This is done in three steps:

- Run our new model alongside gold standard model on the same text
- Compare the results of these runs to generate confusion matrix
- Use confusion matrix to calculate metrics

```python
nlp_gold = nlp
nlp_pred = spacy.load("transitions_ner_model")

nlp_pred.add_pipe('sentencizer')

ent_gold = []
ent_pred = []

doc_gold = nlp_gold(text)
doc_pred = nlp_pred(text)                                                  1

for sent in doc_gold.sents:
  for ent in sent.ents:
    if (ent.label_ == "Transition"):
      ent_gold.append((ent.start_char, ent.end_char, ent.text))

for sent in doc_pred.sents:
  for ent in sent.ents:
    if (ent.label_ == "Transition"):
      ent_pred.append((ent.start_char, ent.end_char, ent.text))

TP = 0
FP = 0
FN = 0

for ent in ent_pred:
  if ent not in ent_gold:
    FP += 1
                                                                           2
for ent in ent_gold:
  if ent not in ent_pred:
    FN += 1

TP = len(ent_pred) - FP
TN = len(doc_gold.ents) - TP - FP - FN

print('TP:', TP, '\t| TN:', TN, '\nFP:', FP, '\t| FN:', FN)
```

Figure 2.6: Code for model assessment

In the first step (block 1) we run both models on the text and collect all of the Transition entities that these models could find

In the second step we calculate values for the confusion matrix. They are as follows:

- FP (false positive): number of non-Transition entities which the trained model classified as a Transition. To calculate this metric, we count entities which are found by our model but are not found by the gold standard model.
- FN (false negative): number of Transition entities which the trained model classified as a non-Transition or didn't detect. To calculate this metric, we do the inverse - count entities detected by gold standard model, but not the trained model
- TP (true positive): number of Transition entities classified as Transition. To calculate this metric, we subtract FP from the total amount of positives (entities detected by trained model)
- TN (true negatives): number of non-Transition entities classified as non-Transition (or undetected). To calculate this, we subtract FP, FN and TP from the total amount of entities (number of entities detected by gold standard model)

2.4 Evaluation and Discussion

To evaluate our model, it's not enough to just use a confusion matrix. Instead we will accuracy, precision, recall and F1 metrics, which are accepted by scientific community and are commonly used to evaluate performance of various algorithms, such as NLP, machine learning and information retrieval.

These metrics are computed as follows:

- Accuracy: shows how accurate our model is in general (that is, how many entries did we correctly label out of all entries)

$$Accuracy = \frac{TP + FN}{TP + TN + FP + FN}$$

- Precision: shows how likely it is for a positive predicted by our model to actually be a true positive (in other words, how many of entities labeled as transitions are actually transitions)

$$Precision = \frac{TP}{TP + FP}$$

- Recall: shows how much out of all positives were detected by our model (that is, how much of true positives were labeled as such, as opposed to false negatives)

$$Recall = \frac{TP}{TP + FN}$$

- F1: F1 is a harmonic mean between precision and recall. In application, it is similar to accuracy, but performs better in case of uneven class distribution (for example, large number of true negatives, which is true for this particular test) or uneven cost of false negatives versus false positives. Overall, it provides more balanced score of overall model performance.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

For the final evaluation, values for confusion matrix are presented in the table 2.1. All the values were calculated based on a 500,000 character long data set created out of scientific paper abstracts.

Table 2.1: Confusion matrix

	Positive	Negative
True	4388	3742
False	283	953

Calculated metric values are provided in table 2.2.

As we want to detect as much of transitions present in the text as possible, we want

Table 2.2: Calculated metric values

	Raw value	Approximated % value
Accuracy	0.8680333119795003	86.8%
Precision	0.9394134018411475	93.94%
Recall	0.8215689945703052	82.16%
F1	0.8765481422293248	87.65%

to minimize false negative metric. We can evaluated how well our model performed on that using recall metric. While the value of 82% is relatively high, based on this metric our model has room for improvement.

This is also shown (though to a lesser degree) by our F1 metric. It is relatively high, and it's specifically applicable to our particular case: there are both uneven class distribution with large amount of true negatives and uneven weights of errors (as we would much rather see false positives, which might be transitions not considered while training initial model, than false negatives, which would mean that we missed a transition that our model should've known about and successfully detected).

Considering the relatively small size of the training data (comprised of approximately 11000 characters), we can conclude that overall the performance of the model was a success. With all of the metrics being greater than 80%, the result is mostly agreed upon to signify a successful model performance (based on the papers reviewed in section "Literature Review").

However, there are still certain ways to improve model performance. For example, we can train it on a larger amount of data or increase initial transition word dataset. Methods to improve model's performance are discussed in detail in the following chapter.

2.5 Conclusion

A lot of current projects require processing raw text in some way. While previously main ways to do this were manual processing and text mining, with the recent rapid development of neural networks, NLP became another viable approach to such tasks. We confirmed that with an existing body of works which successfully implemented NLP for their needs, and developed our own solution to a problem that was not previously covered: detection of transition words in a text.

This model is usable in a variety of different applications:

1. For educational purposes - it can be used as a tool that detects transitional words in a medium such as thesis papers or academic writing in general. As the model is trained on academic texts already, it can also provide a transitional word that will fit certain context both stylistically and contextually. This way, student can either get feedback on chosen transitional words or get a suggestion of one. In both cases, this would provide an opportunity to educate students on academic writing

2. For proofreading assistance - if the user doesn't require suggestions for transitional words, this model can still be used for text analysis. It can provide useful data, such as usage by word (which can, in turn, be used to eliminate potential over-usage of certain words or repetitions)

3. Data analysis - model can also be used to analyze texts for certain trends - for instance, it can be used to detect which transitional words are used more or less often, as well as new emerging transitional words (which can be useful for linguistic research purposes)

Overall, this model successfully performed its function: it detected most of transition words and, even more importantly, successfully detected some tokens that exhibit properties of a transition word while not being in the initial set (one particularly notable example being the token ”/” in constructions such as ’varied with mode/site of delivery’). However, as recall metric denotes that our model missed about 1 in 5 transition words, it would have to be improved in the future. There are several approaches, all of which will be taken in later development:

1. Increase the transitional word data set: if we do that, we can reduce the number of false positive entries by including some of the transitional words that were detected, but weren’t considered while creating initial data set.

2. Increase the amount of training data: model was trained on a relatively small hand-crafted data set. In the final stages of model development, a bigger data set usable for training was found, but was only utilized in the evaluation of the model. Later on, the model can be trained on this data set as well.

3. Employ customized modules: for instance, we can modify POS (part of speech) module and implement it to remove words which couldn’t possibly be a transition word. For example, if we conclude that there are no adjectives in our transition words, we can exclude them as negatives later on, which would improve accuracy of our model.

4. Implement feedback functionality: in certain cases, we can ask user to mark certain transitional words as correct or incorrect. This would be most fitting to use with a low-confidence results. Data collected that way could be used to improve the model as well as potentially creating individual user preferences.

Additionally, following extensions and improvements to the existing program are considered for future development:

1. Develop a GUI: currently, application only has a CLI. One of the first steps to improve its usability would be to introduce a GUI. This way, we can streamline

part of the functionality (for example, providing text to analyze and exporting results) and enable future functionality (such as auto-correction, as described in the next point).

2. Develop transition word proposal functionality: as we already have a model that can predict how fitting a word is to be a transition, we can later on extend the functionality to propose a transitional word based on a given position in sentence and the sentence itself. This, in turn, can be used in several different ways (i.e. automatic correction and word suggestion).

3. Consider Information retrieval (IR) applications: IR could be employed by this algorithm in case of implementing additional functionality. In particular, we can use IR to compare multiple documents when we would offer the most fitting transitional word for a particular case. IR would help to compare submitted text to an existing text, which is either a text previously written by the same author (if our goal is to adhere to a particular author's style) or a document written with a specific style guidelines in mind (i.e. academic writing).

Chapter Transition: Having established this foundational competence in applying traditional machine learning models directly to academic texts, a critical plateau was reached. It was clearly evident that while rigid programmatic extraction is highly effective for classifying specific tokens and structuring text, it is totally insufficient for enacting complex, context-heavy reasoning. Legacy machine learning models lacked native semantic reasoning capabilities. To understand exactly how the global educational domain was attempting to address these limits—and what critical gaps still existed in specialized highly-sensitive sectors like Islamic Jurisprudence—a comprehensive step back was required. Consequently, the following chapter conducts a rigorous, exhaustive Systematic Literature Review (SLR) to map the entire landscape of Educational AI immediately preceding the paradigm shift toward Large Language Models.

Chapter 3

Systematic Literature Review: Generative AI in Specialized Educational Domains

3.1 Introduction

Artificial Intelligence (AI) has emerged as a transformative force in various sectors, and education is no exception.

Our initial hypothesis suggests that AI can enhance education. In this study, we explore the methods through which AI can achieve this enhancement and the potential impact of these methods on educational facilities. These research questions guided our Systematic Literature Review (SLR).

To conduct the systematic literature review, we reviewed previous SLRs related to AI in education. According to the referenced research [16], educational institutions began exploring the integration of AI in education, particularly between 2019 and 2021. Numerous applications have been conducted within these institutions, with one of the most common being the prediction of student performance based on grades. This application can also extend to predicting dropout risks, forecasting student academic performance, and determining student satisfaction with online courses [16] [17].

Furthermore, research has categorized the usage of AI in educational facilities into four main categories [18]:

Firstly, in the context of learning, AI can be leveraged to evaluate student feedback and assignments, generating valuable insights to refine the learning process. Secondly, in the realm of teaching, AI enables the creation of adaptive instructional strategies, customizing the educational experience according to each student's unique needs and learning methods.

The third application is in assessment, where AI enables the integration of automatic grading systems, making the assessment process more efficient and providing prompt feedback to students. Lastly, AI plays a crucial role in administration, enhancing the efficiency of management systems in educational institutions and simplifying a wide range of administrative duties.

Machine Learning (ML) techniques have been applied with positive effects in higher education, particularly in predicting student performance, enhancing the quality of education, and reducing student attrition. ML algorithms are being used to assess students and provide continuous feedback to teachers, both in a classroom and blended learning setting. Several studies have explored ML-based approaches to combat cheating and ensure the academic integrity of student assessments [17].

In their study [16], four major functions of AI applications in online higher education were identified:

Firstly, they enable predictive analysis, accurately forecasting student performance and satisfaction levels in both online and traditional classes. Secondly, they facilitate personalized resource recommendations. AI-based systems recommend relevant learning resources to online students, taking into account their preferences, interests, prior knowledge, and memory capacities.

Thirdly, automatic assessment is another key function. Researchers have developed automated essay evaluation systems, providing immediate assessment, feedback, and automated grading in online English learning platforms.

Finally, AI applications significantly enhance the learning experience. For instance, through the creation of Virtual Reality (VR) tools integrated with AI technology, especially in history learning. These VR tools allow students to fully engage with virtual environments, emulating real cities, where they can learn by exploring and interacting with virtual inhabitants.

The research conducted in this field has yielded promising results and has the potential to significantly impact educational facilities. These researchers have inspired our work in this SLR to delve deeper into enhancing the quality of education. The structure of this paper is as follows: We present our search protocol in Section **??**. Subsequently, we address our research questions in Section **??**. A summary of our findings, highlighting the significance of this research, is provided in Section **??**. Lastly, we discuss the limitations of our study, potential threats to validity, and the review assessment in Section **??**.

3.2 REVIEW PROTOCOL

This study was conducted using a systematic literature review methodology, following the guidelines set forth by the PRISMA checklist. Additionally, we adhered to the research protocol outlined in [19]. We initiated our study by formulating precise research questions that served as the foundation for our investigation.

Subsequently, in alignment with these questions, we crafted a comprehensive search query. This detailed query was employed across three distinct digital libraries, thereby ensuring access to a wide range of potential sources for inclusion.

Upon collecting the resources, we applied stringent inclusion and exclusion criteria to filter the literature. This rigorous screening process ensured that only the most relevant and high-quality papers were included for further analysis.

Finally, after meticulous evaluation and filtering, we arrived at a final selection of key papers for thorough reading and review. These articles formed the cornerstone of our study, providing in-depth insights that effectively addressed our research questions.

3.2.1 PRISMA Checklist

The full PRISMA checklist can be found in Appendix 3.7.

3.2.2 Research Questions

Our research was based on the hypothesis: Can we use AI to enhance education, and to what extent can these applications impact our educational facilities? Based on this hypothesis, we formulated the following research questions:

RQ1. What are the use cases in which AI is employed to gain insights into enhancing education? Additionally, what AI algorithms and techniques are commonly used for this purpose?

RQ2. What are the metrics commonly utilized to assess the quality of both online and offline education? Furthermore, what factors are considered in the evaluation of these metrics?

RQ3. What is the impact of AI on the quality of education in both online and offline teaching formats?

In this study, we aim to explore the role of artificial intelligence (AI) in enhancing education and its impact on educational quality in both online and offline teaching formats. Our first research question (RQ1) focuses on identifying the various use cases of AI that have been employed to gain insights and improve educational practices. We will investigate how AI algorithms and techniques are utilized in educational settings to enhance teaching and learning experiences. RQ2 examines the metrics commonly used to assess the quality of both online and offline education. We will investigate the factors considered in evaluating these metrics, such as student engagement, learning outcomes, instructor effectiveness, and technological infrastructure. Finally, RQ3 aims to understand the overall impact of AI on the quality of education in both online and offline formats. We will examine the effects of AI implementation on student performance, engagement, and satisfaction, as well as the potential benefits and challenges associated with AI integration in educational environments. Through this research, we seek to gain valuable insights into the use of AI in education and its implications for educational quality.

3.2.3 Search process

The databases we used include: Scopus, IEEE Xplore, and ACM Digital Library. W

Search Queries & initial results:

To ensure that the search results for measuring the quality of higher education courses using AI-related technology we divided the query into three sections 1)AI- technologies 2)Quality measures 3)higher education synonyms. For example, we can use search terms such as ”artificial intelligence”, ”machine learning”, ”natural language processing”, or ”deep learning” for the AI technologies. For the quality measures section, some possible search terms could be ”course quality”, ”learner satisfaction”, ”teaching effectiveness”, or ”MOOC satisfaction”. To explore higher education in both online and offline contexts, we can use synonyms such as ”post-secondary education”, ”tertiary education", "university education", or "college education". Putting it all together, a potential search query format could be: "AI technology AND quality measures AND higher education synonym".

("Text mining" OR "sentiment analysis" OR "opinion mining" OR "NLP" OR "AI" OR "supervised machine learning algorithm" OR "sentiment analysis" OR "AIEd" OR "Artificial Intelligence in Education" OR "machine learning" OR "neural network" OR "deep learning" OR "NLP" OR "Natural language processing" OR "unsupervised machine learning algorithm" OR "text analysis" OR "Artificial Intelligence in Education") AND ("course quality" OR "user satisfaction" OR "learner satisfaction" OR "student satisfaction" OR "teaching quality" OR "course satisfaction" OR "MOOCs satisfaction" OR "MOOC satisfaction") AND ("online education" OR "online learning" OR "e-learning" OR MOOC* OR SPOC* OR "blended learning" OR "higher education" OR "online higher education" OR "undergraduate education" OR "graduate education" OR "university" OR "offline education" OR "postgraduate" OR "post-secondary education" OR "tertiary education" OR "higher education", "Online courses", "Online course" OR "college education")

ScienceDirect Query: since in scienceDirect we can't use advanced search, below is the query we used. "AIEd" OR "Artificial Intelligence in Education" OR "machine learning" OR "deep learning" OR "Natural language processing" OR "education" OR MOOCS in abstract "course quality" OR "user satisfaction" OR "learner satisfaction" OR "student satisfaction" OR "teaching quality" OR "course satisfaction" OR "MOOCs satisfaction" OR "MOOC satisfaction"

WE haven't used another query because the results from the mentioned query produced many results. as we will discuss further in the up coming section.

In our Systematic Literature Review (SLR), we've amassed a substantial corpus of papers from three renowned digital libraries. A total of 424 papers were retrieved from

ScienceDirect, marking the highest contribution. This was closely followed by 343 papers from the Association for Computing Machinery (ACM) database. Meanwhile, the IEEE Explore database yielded a robust selection as well, with 316 papers. These figures represent our comprehensive and far-reaching exploration of the available literature on the topic, indicating the thoroughness of our SLR. The comprehensive breakdown of these results can be found in the detailed SLR figure we've compiled. Everything is illustrated in figure 3.1

1. **ScienceDirect (424 papers)**
2. **IEEE Explore (316 papers)**
3. **ACM (343 papers)**

3.2.4 Inclusion and exclusion criteria

We formulated Exclusion Criteria (EC) and Inclusion Criteria (IC). EC and IC help deciding what research to include to our SLR.

Inclusion Criteria (EC) as follow:

IC1- Research must report the application of AI in enhance or measure the quality of education.

IC2- Research must reported as empirical research to demonstrate how they used AI algorithm in measuring course quality.

IC3- Research must report what AI algorithms/methods/system they used conducting the research.

IC4- Research must report the metrics they used to measure the quality of the course.

Exclusion Criteria (EC) as follow:

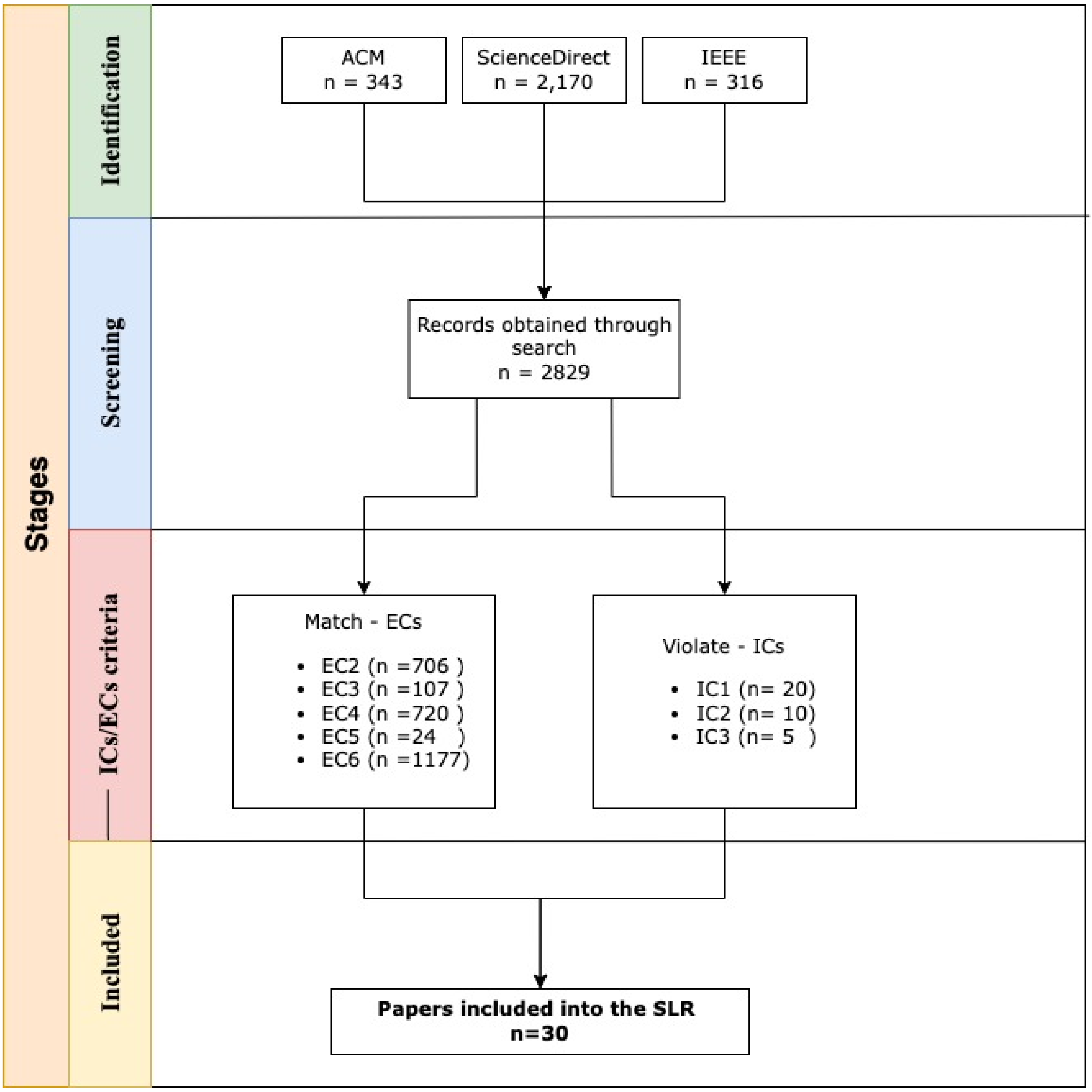

Figure 3.1: SLR diagram

Table 3.1: Table 1 - Databases Pros and Cons

Database	Pros	Cons
IEEExplore	peer-reviewed	-Require subscription -Some papers can't be downloaded directly
ScienceDirec	Filter by CS categories	-No advanced search
ACM	peer-reviewed	Access to papers only through subscription -Some papers can't be downloaded directly

EC1- Research must be written in English

EC2- Research must be related to software engineering or computer science

EC3- Research from conference proceedings, book chapters, magazines, news, and posters, or grey literature are excluded.

EC4- Research must be published from 2013 to 2023.

EC5- Research Full-text is available.

EC6- Research is not duplicated

3.2.5 Primary Clustering

The main advantages and disadvantages of the databases chosen to conduct the search are presented in Table 1. By examining the information presented in this table, it is possible for the reader to determine that all of the databases used are reliable and of high academic quality.

Figure 3.2 displays how the papers are distributed across the selected databases, while Figure 3.3 shows their distribution across publishers. We found a significant representation of papers from different digital libraries. IEEE Xplore Digital Library stands out as the most utilized platform, hosting 18 out of the total papers reviewed. The Association for Computing Machinery (ACM) Digital Library follows with 9 papers. The ScienceDirect (SD) platform, although less represented, contributed 3 papers to our review. This distribution underscores the dominance of IEEE Xplore and ACM Digital Library as primary sources of research related to AI in education, with ScienceDirect also serving as a notable resource.

We analyzed papers from several different publishers. The majority of these papers, 18 in total, were published by IEEE. The Association for Computing Machinery (ACM) was the second most common publisher, accounting for 9 papers. Other publishers such as Computers & Education, Computers in Human Behavior, and Procedia Computer Science each had one paper included in our review. These two figures provide evidence that the papers included in our final selection are sourced from reputable computer science conferences or highly ranked journals.

Furthermore, Figure 3.4 exhibits the distribution of papers by year of publication. It's interesting to observe the frequency distribution of the years these papers were published. The most common publication year was 2019, with a total of 8 papers. This was followed by 2021, which accounted for 6 papers. The years 2020 and 2022 each contributed 5 papers to our review. Earlier years such as 2016 and 2018 were represented by 2 papers each. Lastly, the years 2015 and 2017 were the least common, with only one paper each being published in those years. This distribution provides an overview of the recent trends in research on AI applications in education.

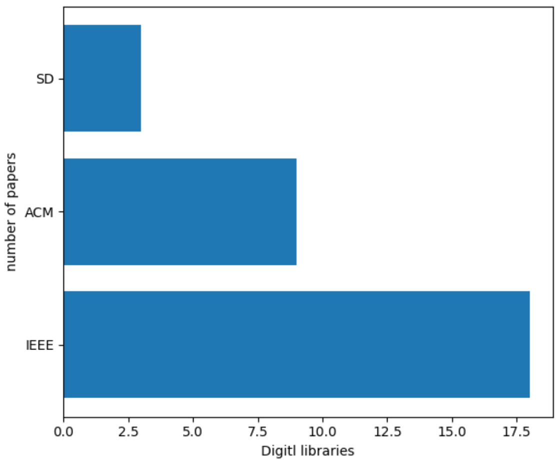

Figure 3.2: Databases distribution

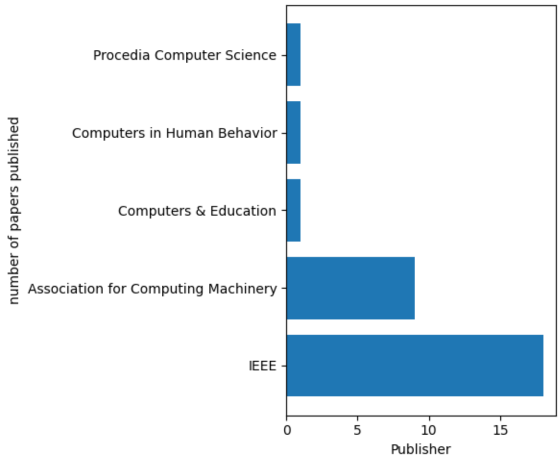

Figure 3.3: Publishers distribution

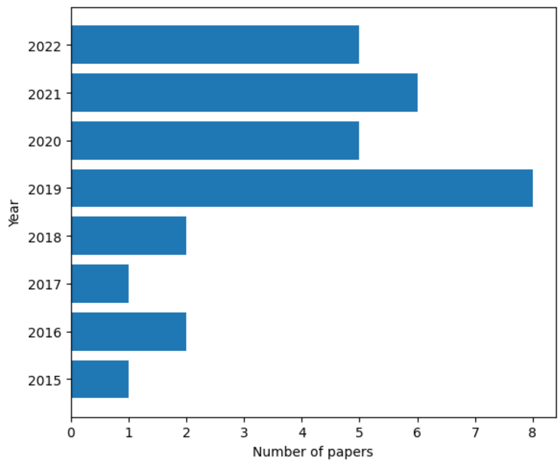

Figure 3.4: Years distribution

3.3 Results and Discussion

3.3.1 RQ1. What are the use cases the AI is used to gain insight to how enhance the education? and what are the AI algorithms/techniques used for such purpose?

After conducting a thorough review of the papers, we have identified five distinct use cases for the research:

1. **Use Case 1-Trio-Techniques Text Analysis for Focused Educational Feedback:** papers which used Sentiment Analysis, Topic Modeling, and Text Clustering Combined

2. **Use Case 2 - Duo-Technique Text Analysis for Focused Educational Feedback:** papers which used Sentiment Analysis and Topic Modeling Together or Sentiment Analysis and Clustering Together

3. **Use Case 3- Solo-Technique Text Analysis for Specific Educational Enhancements:** papers which used Sentiment Analysis alone, Text Clustering alone, or Topic Modeling alone

4. **Use Case 4- Neural Network-Based Forecasting for Educational Outcomes:** Neural Network for Quantitative Data Prediction

5. **Use Case 5 - Computer Vision for tracking emotions:** paper which used Computer Vision approach.

Among these use cases, [20] falls under **Use Case 1** they used Sentiment Analysis to see sentiment values related to course ratings, difficulty levels, and learning duration. Moreover, Emotion Analysis is executed on student feedback, providing an intriguing layer of emotional context to the raw data. The paper also utilizes Topic Modeling to categorize and analyze the key themes present in student feedback. The findings are effectively visualized using color-coded circles to denote sentiment, aiding in data interpretation. This combination of AI methodologies offers a comprehensive approach to understanding and enhancing educational experiences based on students feedback.

Use Case 2, [21] used text Classification to distinguish between a variety of topics including (Content, Technology, Interaction, Pace, Experience), then calculated sentiment analysis for each of the classified topics. Lastly, they did clustering to see which are the hot topics that the students talk about the most, and in which sentiment—positive or negative. In [22] The researchers in the paper employed a systematic approach to analyze satisfaction versus comments in their dataset. First, the data processing, which included converting to lower case, removing stop words, punctuation, The researchers then used Latent Dirichlet Allocation (LDA) for topic modeling. Comments containing common conjunctions were split into separate comments to avoid the confusion that can arise when a single sentence carries both positive and negative sentiments. For instance, a comment might state, ”the pictures are good but the tutorials are bad”. AFINN was employed for sentiment analysis, providing a range between -5 and +5 for more precise sentiment detection. Finally, the researchers utilized visualizations to display sentiment related to specific topic or average satisfaction. The latter was calculated by summing the satisfaction of all comments relating to a particular aspect, then dividing by the total number of such comments. This meticulous approach ensures more accurate and nuanced data interpretation.

Falls under **Use Case 3**, utilizing Topic modeling alone the paper [23] presents a comprehensive methodology for analyzing MOOC data on computer courses. Initially, data was collected through web crawling of MOOC platforms. The subsequent data processing phase involved steps such as data de-duplication, Jieba participle, and removal of deactivated words like ”very good” and ”but”.

They utilized Latent Dirichlet Allocation (LDA) to extract topics of user interest, with eight topics being selected from the LDA topic modeling. Visualization of these results was done using pyLDAvis, enhancing interpretability of the findings.

Additionally, a word cloud was generated featuring the top 100 words with the highest frequency of occurrence, providing an immediate visual summary of the most discussed subjects. This combination of techniques serves to effectively explore and display key areas of user attention within the computer courses.

falls under **Use Case 4**, paper [24] aims to create a robust, scientific system to evaluate teachers' quality using multiple evaluation indices such as teaching attitude, course content, and effectiveness, among others. These indices are inputs for secondary systems, whose outputs feed into a neural network, ultimately determining the final teaching quality evaluation.

Each secondary system employs a three-layer neural network, with the weights and thresholds determined through sample training. The comprehensive evaluation system uses a similar three-layer Backpropagation (BP) neural network, taking outputs from each secondary system as inputs.

The final output is divided into five grades (excellent, good, medium, pass, fail) corresponding to specific score ranges. Thus, the system provides a detailed, layered, and reliable evaluation of teaching quality.

Lastly, falls under **Use Case 5** The paper focuses on analyzing the relationship between learners' facial expressions and teaching effectiveness in an online environment. It establishes a real-time facial expression recognition system that tracks and analyzes the learners' state, assisting teachers in evaluating the effectiveness of online teaching and controlling the teaching process.

Through observing the changes in learners' facial expressions, the paper aims to understand their emotional state and thereby evaluate the impact of online learning. This involves processes like face detection, extraction of facial expression features, and the application of facial expression classification algorithms. The study utilizes Convolutional Neural Networks (CNN) in TensorFlow to achieve these objectives.

It is worth mentioning the work of Palaute et al. [20], which investigates sentiment and emotional analysis to gain insights into students' mental health. Another relevant study by SUFAT [25] focuses on extracting suggestions from student feedback comments to enhance the overall quality of education.

In conclusion, all of these use cases have a direct impact on enhancing the quality of education through data analysis, whether it involves quantitative or qualitative data. Table 3.2 in our paper provides a distribution of these use cases across the reviewed papers. Please note that this table assumes the use cases are mutually exclusive, meaning that a paper fitting into Use Case 1 cannot be listed under Use Case 2 or 3, and so on.

Table 3.2: Use Cases and Paper References - QDP: Quantitative data prediction, SA: Sentiment analysis, TM: Topic modeling, TC: Text clustering, CV: Computer vision, UC: Use case

Use Case	**Paper References**
UC-1: SA+TM+TC	[26]
UC-2: SA + TA OR SA + TC	[20], [22], [27], [25], [28], [29],[21]
UC-3: SA OR TA OR TC	[30], [23], [31], [32], [33], [34], [35], [36]
UC-4: QDP	[37], [38][39], [40], [24], [41], [42], [43], [44], [45], [46], [47], [48], [49]
UC-5: CV	[50]

The analysis of the findings reveals a diverse range of AI techniques and applications employed in measuring and enhancing the quality of education illustrated in figure 3.5. Among these, sentiment analysis emerges as the most frequently utilized technique, appearing 26 times in the findings. By analyzing and interpreting the sentiment expressed in educational data, researchers gain valuable insights into students' attitudes, perceptions, and emotions. Text classification, text clustering, and topic modeling are also prominent techniques, with varying frequencies of occurrence. These techniques enable researchers to organize, group, and uncover latent themes within educational texts, providing a deeper understanding of the content and underlying concepts.

Quantitative data prediction appearing nine times in the findings. This technique allows researchers to utilize quantitative data to predict various outcomes in education, such as dropout rates, student performance metrics, and satisfaction scores. By leveraging predictive models, educators and policymakers can gain insights into trends and patterns that impact the quality of education

Feature selection, emotional analysis, data processing, classification, also demonstrate their relevance in the field. These techniques contribute to refining data, analyzing emotions, making data-driven decisions.

The findings underscore the diverse nature of AI techniques applied in education and highlight their potential in advancing the quality of education through data analysis, prediction, and classification. They provide researchers and practitioners valuable insights into the prevailing practices and avenues for further exploration in the quest for improving educational outcomes.

Figure 3.3.1 represents the frequency of various AI techniques used in the field of education. Among the techniques, 'SVM' (Support Vector Machine) is the most commonly employed, appearing 6 times. 'Scikit-learn', 'BP NN' (Backpropagation Neural Network), and 'MATLAB' follow closely with frequencies of 4, 6, and 3 respectively. Other techniques such as 'NLTK', 'Spacy', 'NRC', 'Syuzhet', 'CNN', 'TF-IDF', 'n-gram', 'Naïve Bayes', 'AFINN', 'Random Forest', 'K-Means', and 'LDA' also find notable usage. These AI techniques showcase the diversity of approaches employed in the analysis and processing of educational data, enabling researchers to extract meaningful insights and improve the quality of education through advanced machine learning algorithms and natural language processing.

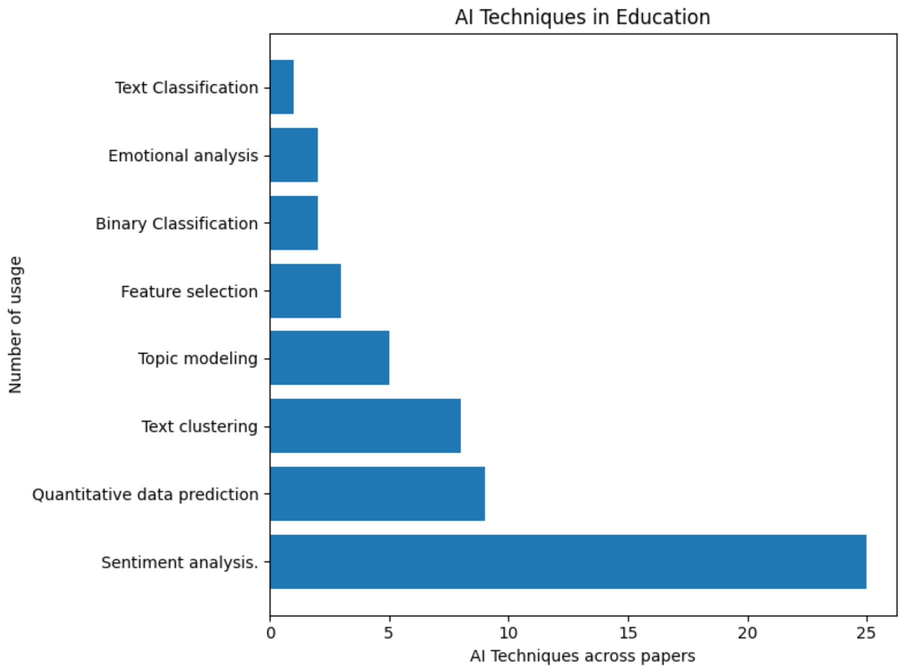

Figure 3.5: AI techniques used across papers

Lastly, we noticed a varying distribution across different types of educational settings ilustrated in figure 3.6. A substantial majority of the papers, 15 in total, pertained to the field of Online Higher Education, indicating a strong focus on the application of AI in virtual tertiary level institutions. Traditional Higher Education was the second most discussed setting, with 9 papers reflecting its relevance. Lastly, MOOCs (Massive Open Online Courses), an emerging platform for online learning, was the subject of 5 papers. This distribution underscores the increasing interest in leveraging AI for improving online education and the broader higher education sector

3.3.2 RQ2. What are the metrics commonly used to assess the quality of online and offline education, what are the performance factors they used to evaluate theses metrics?

Out of the papers we reviewed we noticed that researchers frequently utilize various terms and phrases to express their methods of measuring education quality as metrics.

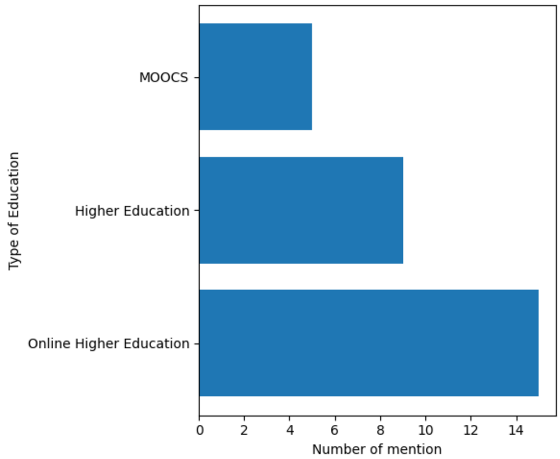

Figure 3.6: Education type distribution

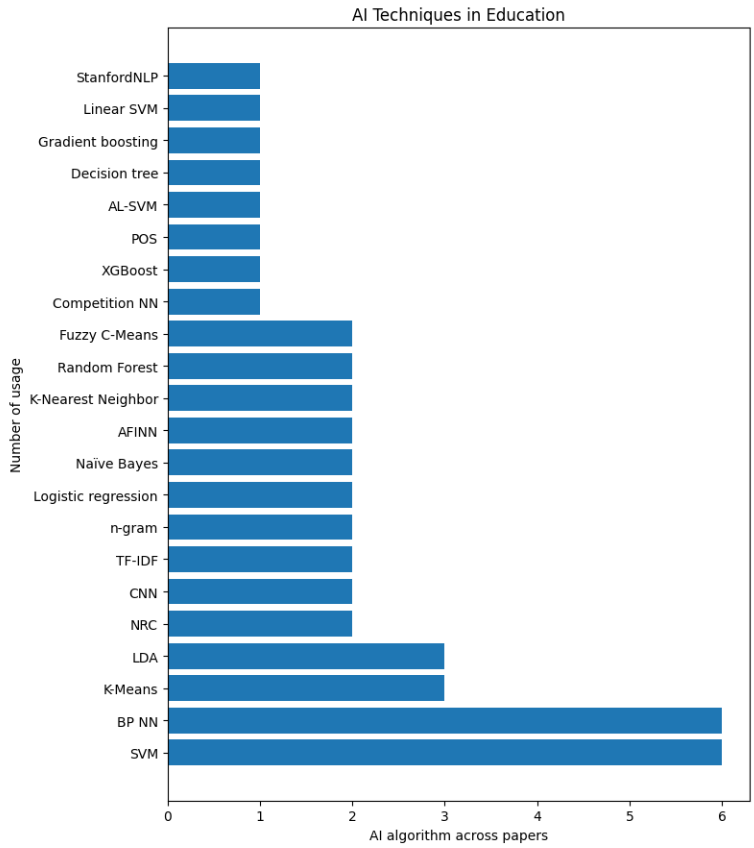

Figure 3.7: AI algorithms used across papers

Despite differences in wording, many of these terms are semantically similar. For instance, two distinct studies might use the metrics ’Teaching quality’ and ’Teaching quality evaluation’; though expressed differently, their meanings are essentially the same. We have therefore categorized these various metrics into four broad groups for clarity and conciseness. These categories are: Teaching Quality and Effectiveness, Teaching Evaluation and Feedback, Learner Satisfaction and Experience, and Course Completion and Retention. The specific metrics that fall under each category are illustrated in Table 3.3.

Table 3.3: Categories and Their Corresponding Metrics

Category	**Metrics**
Teaching Quality and Effectiveness	Teaching quality, Teacher’s teaching quality, Satisfaction with the quality of teaching, Teaching quality of the curriculum, Project-based teaching quality, MOOCs quality, Teaching quality evaluation
Teaching Evaluation and Feedback	Student evaluations of teaching (SET), Teaching evaluating system, Teaching evaluation system TES, Student performance
Learner Satisfaction and Experience	Student satisfaction level, Students satisfaction, Learning experience, MOOC learner satisfaction, Learner satisfaction metric, Students satisfaction score
Course Completion and Retention	Learners’ course completion, MOOCs dropout

In our analysis, we compared the usage of various metrics to discern research trends in AI-enhanced education quality. This exploration helps identify key focus areas and the importance attributed to each metric. It provides insights into prevailing academic interests, potential gaps in the research, and guides future studies for a balanced and

Table 3.4: Categories and Their Corresponding Metrics

Category	Metrics	Paper References
Teaching Quality and Effectiveness	Teaching quality, Teacher's teaching quality, Satisfaction with the quality of teaching, Teaching quality of the curriculum, Project-based teaching quality, MOOCs quality, Teaching quality evaluation	[37], [22][33], [39], [47], [41], [32], [33], [36], [48], [21], [44], [45], [46], [50], [46]
Teaching Evaluation and Feedback	Student evaluations of teaching (SET), Teaching evaluating system, Teaching evaluation system TES, Student performance	[20] , [38] [43], [35]
Learner Satisfaction and Experience	Student satisfaction level, Students satisfaction, Learning experience, MOOC learner satisfaction, Learner satisfaction metric, Students satisfaction score, Course satisfaction	[49], [26], [22], [31], [34], [29], [28]
Course Completion and Retention	Learners' course completion, MOOCs dropout	[49], [27]

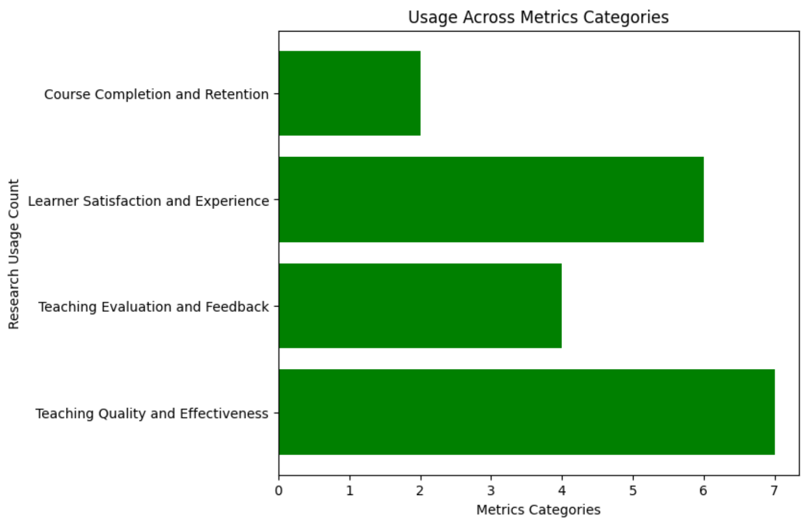

Figure 3.8: Metrics Categories.

comprehensive investigation of the field. The comparison is better represented in figure 3.8. below is the interpretation about the metrics usage we analysed and the complete distrbution of the paper can be found in Table 3.4:

1. **Teaching Quality and Effectiveness:** This category appears to have the highest metric usage with a count of 7. It seems to be the most popular category, which may suggest that most of the focus in the data is on evaluating the effectiveness of the teaching quality and how to improve it.

2. **Teaching Evaluation and Feedback:** The metric usage for this category is moderately high with a count of 4. This indicates that the evaluation and feedback of teaching is also a critical aspect, though not as prominent as teaching quality and effectiveness.

3. **Learner Satisfaction and Experience:** This category has the second highest usage with a count of 6. It suggests that a significant emphasis is also given to the satisfaction and experience of the learners. This might be due to the recognition of its impact on overall learning outcomes.

4. **Course Completion and Retention:** This category has the lowest usage with a count of 2, indicating that less focus is put on course completion and retention compared to the other categories. This might be because other aspects of teaching and learning, such as teaching quality and learner satisfaction, are perceived to be more influential on overall education quality.

lastly, we should mention that papers[42] and [30] didn't mention any metrics.

Based on the usage counts, most of the focus is on 'Teaching Quality and Effectiveness' followed by 'Learner Satisfaction and Experience'. 'Teaching Evaluation and Feedback' and 'Course Completion and Retention' also play roles but are less emphasized in the current data. It could be beneficial to consider whether a more balanced focus across all these categories might lead to a more comprehensive understanding of the quality of education.

The second phase of our metric analysis involves examining the factors utilized in the computation of these metrics. Based on the findings, we can structure and categorize the various factors of educational quality into four main groups listed below. Each of them mention categories encompasses several factors, all of which are detailed in Table 3.5:

1. **Educator Performance Metrics:** This group focuses on factors related to teaching quality, style, and effectiveness.

2. **Student Engagement and Performance Metrics:** This group measure student behaviors and achievement, which also indirectly reflect the effectiveness of teaching.

3. **Course and Material Quality Metrics:** This group involves the structural components and delivery of the course itself.

4. **Environmental and Institutional Metrics:** This group is based on factors that are less directly related to the teacher-student interaction, but still have an influence on educational quality.

In conclusion, we analyzed the metrics commonly used to assess the quality of online and offline education. We categorized these metrics, compared their usage, and explored the performance factors associated with them. We also grouped the factors into three main categories to provide a structured overview. Our aim was to gain insights into the evaluation methods used in assessing educational quality.

Table 3.5: Metrics Factors

Educator Performance Metrics: Teaching attitude, Teaching method, Teaching content, Teaching ability, Teaching style and ethics, Faculty interaction, Faculty feedback, Faculty preparation, Classroom delivery
Student Engagement and Performance Metrics: Student satisfaction, Student attendance, Student marks, Student achievements, Age, Course rating, Would take again, Login frequency, intervals, and durations
Course and Material Quality Metrics: Course difficulty, Course content, Course description, Courseware design, Learning management, Interactive program, Course structure and contents, Technology and Support, E-learning pace, Overall experience, Course schedule, Course major, Course duration, Time investment required
Environmental and Institutional Metrics: Class style, Teaching staff, School conditions, Quality of students, Campus culture, School spirit, Geographic and psychological environment

3.3.3 RQ-3 What is the impact of AI on the education quality for both online and offline teaching formats?

In our review of the selected papers, we examined the educational elements impacted by AI techniques and noticed a degree of similarity between these elements. Consequently, we have categorized them into the following clusters:

- **Cluster-1: Feedback Analysis and Course Design** This cluster includes papers which focus on analyzing student feedback and modifying course design accordingly.
- **Cluster-2: Teaching Quality and Effectiveness** Papers in this cluster are centered around evaluating teaching quality and effectiveness, including course content, teacher's effectiveness, and teaching strategies.
- **Cluster-3: Course Assessment and Student Satisfaction** This cluster focuses on course assessment, student satisfaction, and the factors influencing them.
- **Cluster-4: Course Duration, Difficulty and Completion Rates** This cluster includes papers that focus on course duration, difficulty, completion rates, and

dropout rates.

In the ”Cluster-1: Feedback Analysis and Course Design”, Palaute et al. [20] introduced a tool designed to extract and analyze student feedback. This tool identified specific areas requiring improvement within the course, thus profoundly influencing the design and faculty-related aspects of the course. Similarly, SUFAT, an analytics tool employed by [25], analyzed qualitative feedback comments from students. This information directed instructors toward areas of concern, guiding informed decisions for effective course modifications, thereby enhancing teaching effectiveness and decision-making processes.

In the ”Cluster-2: Teaching Quality and Effectiveness”, the study by [32] elucidated students' perceptions of teaching attitude, course content, methods, and effectiveness, consequently identifying areas for improvement. This insight considerably impacted teaching effectiveness and course content. Moreover, the unique approach adopted by [47] fused student characteristics and feedback to evaluate teaching quality in higher education institutions, thereby influencing the quality of teaching and course evaluations directly.

Moving on to the ”Cluster-3: Course Assessment and Student Satisfaction”, [26] proposed a data mining technique to predict and enhance student satisfaction, allowing early interventions to mitigate educational issues, especially during crises such as COVID-19. This intervention directly improved student satisfaction and the efficacy of online learning. Furthermore, [31] found that aspects such as course assessment, instructor presentation, and course content significantly influenced learner satisfaction, whereas course video and interaction did not significantly impact it.

Lastly, in the ”Cluster-4: Course Duration, Difficulty, and Completion Rates” cluster, the research conducted by [37] identified curriculum quality, class hours, and curriculum difficulty as the most significant factors affecting classification, directly impacting curriculum quality, course duration, difficulty, and completion rates. Simultaneously, [27] applied classification models to course attributes, revealing correlations between course completion rate, duration, and difficulty.

In conclusion, the research papers provided key insights into how AI techniques can be utilized to improve various elements of education. The effectiveness of AI tools in enhancing course design, teaching quality, student satisfaction, and understanding of course duration, difficulty, and completion rates was evident. For a comprehensive understanding and in-depth analysis of the individual contributions of each research paper, please refer to Table 3.6.

3.4 SYNOPTIC SUMMARY AND SIGNIFICANCE OF RESEARCH

In our comprehensive review, we observed multiple use cases of AI application intended to enhance education, in both online and offline contexts. After an extensive study, we identified a pattern grouping these use cases together:

1. Dual-Technique Sentiment Analysis for Focused Educational Feedback.
2. Single-Technique Text Analysis for Specific Educational Enhancements.
3. Neural Network-Based Quantitative Forecasting for Educational Outcomes.
4. Computer Vision for Tracking Emotions.

A detailed explanation of these groupings is available in Section 3.3.1. Interestingly, sentiment analysis was found to be the most frequently utilized AI technique, with many researchers aiming to analyze feedback comments for more profound insights.

Second, several metrics were employed to measure teaching quality. We noticed

Table 3.6: AI-Enhanced Education Evaluation

Cluster	**Improved Educational Element**	**Paper References**
Feedback Analysis and Course Design	Course Design & Faculty, Course modifications, teaching effectiveness, instructor's decision making, Faculty interaction, course structure, Course development, student feedback.	[20], [25], [21], [33], [22], [36]
Teaching Quality and Effectiveness	Curriculum quality, course duration and difficulty, completion rates, Teaching quality, course evaluations, Teaching effectiveness, course content, College teaching quality, Chinese International Education quality.	[37], [47], [32], [45], [41]
Course Assessment and Student Satisfaction	Course Duration & Difficulty, Student satisfaction, online learning efficacy during crises, Student's college experience, department reputation, course ratings, Course assessment, instructor attributes, course content, Course assessment, learner feedback, Course satisfaction, student willingness.	[27], [26], [38],[28], [31], [29], [34]
Course Duration, Difficulty and Completion Rates	Curriculum quality, course duration and difficulty, completion rates, Course Duration & Difficulty, Course dropout rate, learning habits.	[37], [27], [49]
No mention for impact	N/A	[39], [40], [30], [24], [48], [42], [43], [44], [35], [46]

these metrics can be clustered based on their utility and their area of impact. The identified clusters are:

1. Feedback Analysis and Course Design.
2. Teaching Quality and Effectiveness Papers.
3. Course Assessment and Student Satisfaction.
4. Course Duration, Difficulty and Completion Rates.

We further analyzed the factors utilized within these metrics. Details of these findings are presented in Section 3.3.2.

Third, we examined the impact of AI techniques used in reviewed papers, and the educational elements impacted by these techniques.Details of these findings are presented in Section 3.3.3. The identified clusters of these elements are:

- Feedback Analysis and Course Design.
- Teaching Quality and Effectiveness Papers.
- Course Assessment and Student Satisfaction.
- Course Duration, Difficulty and Completion Rates.

Lastly, not many researcher created a tool that the community can use except researches:[20], [25], [36], [29], and [21].

3.5 LIMITATIONS, THREATS TO VALIDITY, AND REVIEW ASSESSMENT

3.5.1 Limitations

1. Our study confines itself to research from the past decade, introducing a potential limitation in acquiring comprehensive knowledge. However, precedent Systematic Literature Reviews (SLRs) in this area, such as the one by [17], observed that the majority of relevant studies were published between 2016 and 2019. Our analysis of publication years resonated with these findings illustrated in figure 3.4, suggesting a concentration of research activity during this period. This observation gives us confidence that our 10-year limit, while potentially excluding some earlier work, is unlikely to significantly impact the thoroughness or relevance of our findings.

2. The search was carried out utilizing a limited number of databases, namely IEEExplore, ScienceDirect, and ACM Digital Library. Some readers may argue that we should have considered additional databases for our searches, such as Google Scholar.We observed that Google Scholar often simply compiles papers from other databases, sometimes with no filter for academic quality. This leads to low-quality conference proceedings and questionable gray literature often being found through Google Scholar. Although that was our selection of database but we strongly believe that this selection helped us to insure solid academic findings

3. While conducting this systematic literature review, we acknowledge the potential shortcoming of not considering grey literature. It is common for SLRs to prioritize primary studies over grey literature, which could potentially lead to missing out on relevant findings. However, in order to ensure the quality and rigor of our review process, we made the decision to exclude grey literature from our analysis. While

this may have limited the scope of our review, it allowed us to focus on high-quality studies and provide a more rigorous evaluation of the research in the field.

4. To conclude, it could be contended that the decision to solely include papers written in English, as one of the inclusion criteria in this study, could be considered as highly limiting the scope of research available. Nevertheless, it is important to note that a significant portion of the existing literature in the field is written in English, hence this criterion is not peculiar or rare in the context of our field of research.

3.5.2 Threats to Validity

This section of the paper aims to address any possible biases that may have influenced our research and the results we obtained. We will outline the measures we took to prevent the occurrence of such biases.[51]

1. **Retrieval bias** - In order to avoid any potential bias, we carried out extensive searches across three different databases. We also made sure to select only high-quality articles, and we only included works that were relevant to our research topic. although we might be biased and limited the search by last 10 years of publication.

2. **Publication bias** - Our research was based on high-quality, peer-reviewed articles that were published in credible academic journals or distinguished conferences. We aimed to cover a range of analysis levels and experimental protocols, to ensure that we did not solely report statistically significant results.

3. **Inclusion criteria bias** - To prevent bias, we endeavored to develop the most appropriate and inclusive criteria for our subject matter, adhering to the highest standards of our field.

4. **Selector bias** -we thoroughly reviewed all the papers that passed our exclusion criteria to ensure the validity of our selection. Some papers were deemed obviously irrelevant to our research, and were therefore excluded. Other papers were selected based on the aforementioned inclusion criteria, and we took care to read through the methodology, discussion, and results of each selected paper to ensure that we obtained the best results that matched our findings. This rigorous process was adopted to guarantee the quality and accuracy of our research.

3.5.3 Review Assessment

The final stage of evaluating our research findings involves an overall assessment of the quality of our work. As suggested in [52], we have created a list of questions that can be used as a benchmark for assessing the quality of an SLR. These questions are provided below for reference.

1. **Were the inclusion and exclusion criteria based on reasonable and objective grounds?** Before starting the search process, we mentioned all the criteria in the methodology. The criteria we chose are in agreement with the ones usually used in the field and are appropriate for the topic of our research. Therefore, we consider the inclusion/exclusion criteria we utilized to be rational.

2. **Was the search process comprehensive and covered all potentially relevant papers?** We conducted our searches using trustworthy bibliographic databases and collected papers from highly-regarded academic journals and reputable conference proceedings. Based on this approach, we can state that the papers chosen for inclusion in our final record are indicative of the field.

3. **Was there a quality review of the selected papers?** We created a measurement standard to evaluate the quality of the papers. As a matter of fact, we demonstrated that the papers included in the record possessed a high level of quality. Hence, we

can affirm that the conclusions drawn from our systematic literature review were dependable and precise.

3.6 conclusion

In concluding our comprehensive research, we have established key findings and potential directions for future investigation in the field of AI application in education, informed by the analysis of 30 diverse papers through a systematic literature review (SLR).

Firstly, regarding the use-cases of AI in education (RQ1), four prominent approaches emerged. These included dual-technique sentiment analysis, single-technique text analysis, neural network-based quantitative forecasting, and computer vision applications. Sentiment analysis emerged as the most prevalent technique, focusing primarily on mining and interpreting feedback comments to provide more comprehensive insights, potentially leading to more focused and effective educational interventions. However, it's clear that these techniques are not mutually exclusive and can be combined to yield richer insights and more nuanced applications in education.

In terms of measuring the quality of education (RQ2), several metrics were identified, falling into clusters of Feedback Analysis and Course Design, Teaching Quality and Effectiveness, Course Assessment and Student Satisfaction, and Course Duration, Difficulty and Completion Rates. These metrics address a range of aspects in both online and offline education, suggesting that the quality of education is a multidimensional construct that requires a multifaceted approach to measure effectively. Each cluster has its own set of factors that contribute to their evaluation, indicating the depth and complexity of what constitutes 'quality' in education.

Thirdly, the impact of AI on educational quality (RQ3) was found to be profound across both online and offline formats. The same clusters as in RQ2 emerged, demonstrating that AI can be utilized effectively across these diverse aspects of educational quality. The ability of AI to analyze feedback and design courses, assess teaching quality and effectiveness, evaluate course satisfaction, and track course duration, difficulty and completion rates, underscores its transformative potential in enhancing educational quality and student outcomes.

Overall, the breadth and depth of AI application in education are vast, and its potential to enhance quality is substantial. As AI techniques continue to evolve and mature, their influence on education is expected to grow, warranting continuous study and refinement. Future research should also seek to address existing gaps and explore new AI techniques to further the understanding and application of AI in education.

3.7 Appendix

3.7.1 Prisma checklist

the Prisma check list can be found in the table 3.7.

3.7.2 Quality Assessment

our evaluating of the paper is out of 3 and the average quality of the papers is 2.53. all the quality of the papers presented in table 3.8. detaile explination of the assessment points are below: every paper will take 1 point out of the three categories below thus accumlating 3

1. **Q1- Does the research provide a detailed description how they used AI technology in their research?**

   (a) 1 - if the research provide full description

   (b) 0 - if the research intentionally hide how they are solving the problem

Number	Location
TITLE	
1	page **??**
ABSTRACT	
2	section **??**
INTRODUCTION	
3	section 3.1
4	section 3.1
METHODS	
5	section 3.2
6	section 3.2.3
7	section 3.2.3
8-15	not applicable
16a	section 3.2
16b	not applicable
RESULTS	
17	section 3.3
18-22	not applicable
DISCUSSION	
23a	sections[3.3.1,3.3.2,3.3.3]
23b	section 3.3
LIMITATION	
23c	section 3.5
CONCLUSION	
23d	section 4.7
OTHER INFORMATION	
24	not applicable
25	missing

Table 3.7: Prisma checklist

2. **Q2- Does the research provide detail on the research design methodology and how the reach the results?**

    (a) 1: authors provided full research

    (b) 0: authors didn't provide deep explanation of their methodology

3. **Q3-Does the research provide an explanation how this techniques impacted the educational institution?**

    (a) 1- if the authors provided explanation of the impact on the institute

    (b) 0 if the authors did not provided explanation of the impact on the institute

Chapter Transition: The Systematic Literature Review conclusively demonstrated that the apex of traditional AI in education natively culminated in evaluative architectures—using neural networks and NLP strictly to measure sentiment, predict dropouts, or classify topics. Generative reasoning was fundamentally absent. However, as the global research landscape aggressively shifted with the advent of Large Language Models, mere pedagogical evaluation gave way to a much deeper challenge: could AI actually ingest, reason, and output definitively accurate, highly complex domain knowledge without hallucinating? To test this, the research explicitly pivoted away from general education and directly into one of the most rigorous, high-sensitivity domains available: Islamic Jurisprudence and Hadith Sciences. The foundation for Chapter 3 is thus laid: moving from merely classifying academic literature to architecting absolute ground-truth verification systems.

Paper References	Rating	Missing Quality
[20]	3	-
[27]	3	-
[37]	2	Missing Q3
[25]	3	-
[39]	2	Missing Q3
[26]	3	-
[40]	1	Missing Q3 Missing Q2
[30]	1	Missing Q3 Missing Q2
[28]	3	-
[47]	3	-
[23]	3	-
[24]	2	-
[41]	2	Missing Q2
[48]	2	Missing Q2
[42]	2	Missing Q2
[43]	2	Missing Q2
[31]	3	-
[29]	3	-
[21]	3	-
[32]	3	-
[33]	3	-
[22]	3	-
[34]	3	-
[44]	2	Missing Q3
[50]	3	-
[49]	3	-
[35]	2	Missing Q2
[45]	3	-
[46]	2	Missing Q3
[36]	3	-

Table 3.8: Quality assessment

Chapter 4

Ahadith AI: Navigating Authenticity and Classification

4.1 Introduction

The study of Hadith holds a significant place in Islamic scholarship, serving as a vital source of guidance and inspiration for millions worldwide [53]. In the pursuit of understanding and enhancing this rich tradition, this research endeavors to contribute to the field of Islamic Hadith, with a particular focus on the revered compilation, *Riyad al-Salihin* [53]. Authored by Imam al-Nawawi, this compilation stands as a cornerstone in Hadith literature, offering a treasure trove of wisdom and teachings for believers.

4.1.1 Objective of the Research

This research aims to enhance the understanding and accessibility of Hadith literature through a series of innovative contributions. The drastic changes in life and education after the COVID-19 pandemic in 2019, along with the rise of AI [54] and it's applications in text [55], have created new opportunities for leveraging technology in religious studies. This study also seeks to integrate AI into Hadith analysis to identify similarities and connections between Hadith and other religious texts [56]. The primary objectives of this research include:

1. **Enhancement of Hadith Accessibility:** Adding Arabic titles to each Hadith using the capabilities of ChatGPT3 & 4 API, thereby facilitating easier navigation and comprehension for Arabic-speaking audiences.

2. **Globalization of Hadith:** Translating the generated Arabic titles into English using ChatGPT 3 API, broadening the accessibility of Hadith literature to English-speaking individuals worldwide.

3. **Structural Analysis:** Separating the Sanad (chain of narration) from the Matin (text) of each Hadith, allowing for a more detailed examination of its historical context and authenticity.

4. **Technological Integration:** Incorporating Hadith, Matin, and titles into an embedding space and vector database, enabling advanced computational analysis and categorization of Hadith literature.

5. **Semantic Search Implementation:** Implementing semantic search techniques, including similarity search, Maximal Marginal Relevance (MMR), and contextual compression, to facilitate efficient retrieval and exploration of Hadith texts.

6. **Practical Application:** Developing an Android application named "Ahadeeh.ai" to showcase the research contributions and provide users with a user-friendly interface for accessing and exploring Hadith literature.

7. **API Development:** Creating a public API using FastAPI to provide access to the compiled Hadith texts and enabling users to conduct searches and inquiries on specific Hadith topics.

8. **Dataset Creation:** Curating a comprehensive dataset on Kaggle containing all the Hadiths from *Riyad al-Salihin*, with separated Matin and Sanad, along with Arabic titles translated into both English and Arabic languages, fostering further research and analysis within the academic community.

Through these objectives, this research seeks to bridge the gap between traditional Islamic scholarship and modern technological advancements, thereby fostering a deeper understanding and appreciation of the timeless wisdom contained within Hadith literature.

4.1.2 What is Hadith

Hadith in islam is everything transmitted from the Prophet Muhammad (Salla Allahu Alayhi Wa Sallam), whether it be his sayings, actions, approvals, traits, personal habits, or biography, whether before or after his mission as profit for Islam. Hadith is the second source of Islamic law after the Qur'an, obedience to the Messenger of Allah is also obedience to Allah SWT. The Prophet's Sunnah came to strengthen, explain and add new laws to the Qur'an. The scholars have paid great attention to preserving the Sunnah.

4.1.3 What is the Riyadh Al-Salihin Book?

The book *"Riyadh al-Salihin"* authored by the late Imam al-Nawawi in 676 AH is distinguished for its uniqueness in its field and subject matter. Imam al-Nawawi, may Allah grant him mercy, compiled approximately 1896 authenticated Hadiths from the most reliable sources of the Prophetic Sunnah. He structured these Hadiths into 17 chapters and further subdivided them into 362 segments. His preference was to gather Hadiths from trustworthy compilations, for example, Sahih collections. [57]

4.1.4 Why Choose Riyadh Al-Salihin

Imam al-Nawawi, may Allah have mercy on him, had a specific purpose in mind when compiling his book. He aimed to create a comprehensive guide encompassing the obligations incumbent upon every accountable person, covering aspects of worship such as prayer, fasting, charity, pilgrimage, and more.

Furthermore, he included prohibitions against vices in behavior such as arrogance, backbiting, slander, and similar actions, along with disliked actions, recommended deeds, and the virtues of deeds. This book is thus suitable for individuals at all levels and backgrounds of knowledge, proving beneficial for both the general public and students of knowledge. Its profound benefits have led to its widespread acceptance and circulation throughout the ages.

In ”Riyadh Al-Salihin,” Imam al-Nawawi committed to mentioning only authentic Hadiths from well-known sources (Sanad or Isnad). If it weren't for the Isnad, people could attribute any statement to the Prophet as they pleased [58]. there are many works attempts to simply categorise Hadith as authentic or not such as [59] and [60].

All the Hadith in this book are considered Sahih, n essence, Hadith are categorized into different groups based on their authenticity, ranging from completely reliable to entirely fabricated. These classifications include Sahih (authentic), Hasan (good), Da'if (weak), and Maudu (fabricated) [61].

Imam al-Nawawi supplemented them with verses from the Noble Quran for relevant topics and provided explanations or clarifications where necessary. [62]

4.1.5 What is Semantic search

Semantic search is “search with meaning” [63] a search technique that aims to understand the intent and context behind a user's query rather than just matching keywords. It utilizes natural language processing (NLP) and machine learning techniques to comprehend the meaning of words, phrases, and the relationships between them. This understanding allows semantic search engines to deliver more relevant results by considering the user's query in context and understanding the underlying concepts.

In contrast, lexical search, or keyword-based search, relies solely on matching keywords between the user query and the indexed content. It doesn't consider the meaning or context of the words, which can sometimes lead to irrelevant or inaccurate results.

In the subsequent sections of this paper, we will present a detailed account of the methodology employed to add titles and implement semantic search. Then, we will discuss the results obtained and the discussions arising from our research. Additionally, we will delve into the development of the Ahadeeh.ai application and the free API to integrate Hadith semantic search.

4.2 Methodology

Two main objectives were in mind: to generate titles in both English and Arabic for every hadith, and to add semantic search capability to all ahadith.

4.3 Getting the Data

Our initial goal was to find the Riyadh al-Salihin book in any digital format so that we could later work with its content. We obtained initial data for our study from the GitHub repository 'hadith-json' by Ahmed Abdul Baset. This repository provided valuable datasets on Hadith texts in JSON format. The URL for the repository is https://github.com/A7med3bdulBaset/hadith-json. We considered two JSON files from the repository: the first file contained a list of Hadiths in both Arabic and English, while the second file listed the chapter names with Ids of the Riyadh al-Salihin book.

We wrote a Python script to read this JSON and combine the two lists by adding the chapter name to its corresponding Hadith. As a result, we obtained a CSV dataset containing all the Hadiths in Arabic and English text.

4.4 Generating Titles

4.4.1 Arabic Titles

The first contribution to the book is to generate titles for every hadith using the ChatGPT API. To achieve this, the following steps were taken:

Firstly, we aimed to add titles to every Hadith, as titles can help people skim through multiple Hadiths faster or find ones they prefer. To do so, we first separated the Sanad (list of narrations) from the Matn (the hadith text) (see Figure 4.1). In Riyadh al-Salihin, Imam al-Nawawi had already removed the higher Sanad chain [57]. For this research, our aim was to make a complete separation of the Sanad from the Matn. Fortunately, the original data we worked with already contained a separated English Matn. We will utilize this later in the semantic search phase of the research.

We wrote a Python script to separate Sanad from Matn and then checked for any errors in our separation, correcting them manually. By the end, we had compiled a list of hadith with only the Matn (just the hadith text).

Secondly, we wanted to generate titles for every Hadith using the compiled list. We integrated with ChatGPT-3 (‘gpt-3.5-turbo-0125‘) to generate the titles, utilizing Few-shot prompting to help ChatGPT generate more suitable titles for ahadith. All prompts in detail can be found in Appendix A (see Figure 6).

After execution, we generated Arabic titles for all 1896 Hadiths. However, upon careful review, we discovered that some of the generated titles were not accurately conveying the Hadith (see Figure 2). As Hadith is sacred for Muslims, we cannot afford to have inaccurate titles like these, so we decided to switch to ChatGPT-4 API to generate the titles.

This time, we didn't use Few-shot prompting, given that ChatGPT-4 is state-of-the-art and more proficient in generating titles than ChatGPT-3. We employed a simple prompt (see Appendix A).

The results were significantly better; see Figure 2 for a comparison between titles generated by ChatGPT-3 versus ChatGPT-4.

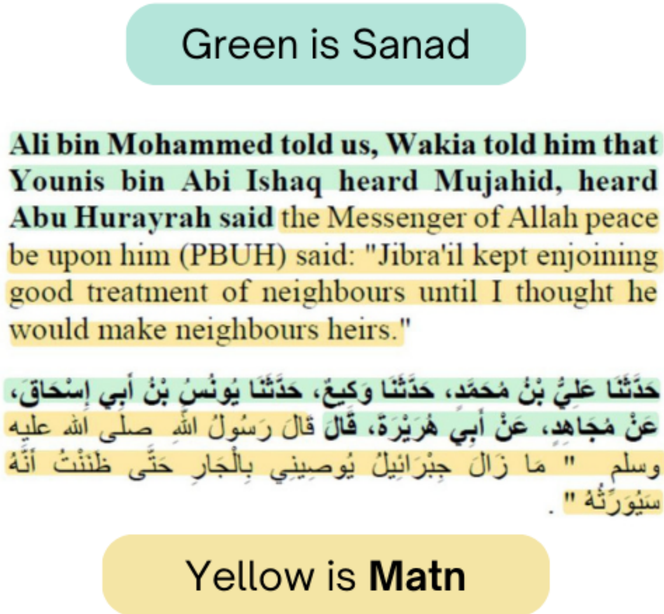

Figure 4.1: Separation of Sanad and Matn

4.4.2 Generating English Titles

This part was relatively straightforward, as we already had Arabic titles. We asked ChatGPT-3 to translate all titles into English, and the results were satisfactory (see Figure 2).

In conclusion, we now have a list containing only Matn and titles in both Arabic and English. All of this data will be used to implement semantic search.

All of the generated titles, along with the separated Matn and Sanad, were later saved into a CSV file together with the original data we found. This file was uploaded in its entirety for anyone to use on the Kaggle website https://www.kaggle.com/datasets/muwaffaqimam/riyad-al-saliheen-book-of-hadith.

4.5 Semantic Search

4.5.1 Creating Embeddings for Hadith Dataset

In order to implement a powerful search mechanism for the Hadith, we decided to employ semantic search using OpenAI embeddings and a vector database with the help of

Langchain library (https://github.com/langchain-ai/langchainjs).

To generate embeddings, we first prepared our dataset by creating a CSV file containing Matn and titles in both Arabic and English. Then, we utilized the CSVLoader from the Langchain library to read this file.

This process helped us divide every row into document objects, which we then used to create a vector database. We filled this database with the embeddings of our dataset. The initial database we worked with was FAISS, and the embedding model used was the OpenAI **text-embedding-3-large**.

This allowed us to create a vector database, which was saved locally. This will later assist us in performing semantic search over our Hadith dataset.

4.5.2 Performing Semantic Search

Now that we have a vector database, thanks to Langchain library wrappers over different vector databases, we can perform semantic search using similarity search and

MMR (Maximal Marginal Relevance). Firstly, we receive a query from the user, then we embed the query using the same embedding model used during vector database creation. After that, we utilize the wrapper methods over the vector databases to retrieve relevant documents either using similarity or MMR algorithms. We also employed a Reranker model to re-rank the retrieved Hadith using the Cohere Reranker model **rerank-multilingual-v2.0** and **ContextualCompressionRetriever** from Langchain. The results are explained in detail in the results section, where we compared all of the aforementioned search methods against multiple vector databases. For the sake of finding the best results, we also conducted experiments to compare the results obtained from three well-known vector databases (FAISS, Qdrant and Chroma). The findings of this comparison are explained in the results section of the research.

Lastly, we have created a mobile application called Ahadeeth.ai `https://play.google.com/store/apps/details?id=com.whilelearn.ahadeeth.ai`: Riyadh al-Salihin Book, which demonstrates our results. We also developed a free API that anyone can use to search for Hadith or obtain them. Both points will be discussed in detail in the App and API development section of the research.

Results

The main results of this research were generating titles for every Hadith in both Arabic and English languages. The second result was incorporating all of the Hadith into a semantic search ability, where users can search Hadith semantically instead of using lexical-based search, which is more powerful.

We also created a mobile application on Android called Ahadeeth.ai, where users can utilize the results of this application by quickly skimming Hadith titles and using semantic search to find specific Ahadeeth. At the time of writing this paper, we had over 200 users and an average of 20-23 app sessions per week. More details about app creation and API development will be discussed in the app development section.

To ensure the delivery of the best results regarding Hadith AI, we decided to conduct comparisons between multiple technologies and seek expert opinions to determine which set of returned Hadith will be considered most correct.

4.5.3 Comparison

In this research, we compared the results against vector databases, and then against a Reranker.

Comparison between vector databases

The comparison we ran was between FAISS, Chroma, Quadrant retrievers, and Cohere reranker. We chose these libraries for their popularity and the number of GitHub stars they have. The initial comparison involved using similarity search. For example, we fed the query "money" or ("Al-Mal" in Arabic), then used all three vector databases as retrievers using similarity search. We recorded the first 5 results and compared them with each other. To ensure a fair comparison, we sought an expert opinion (Sheikh) who knows the Hadith better and can determine which set of returned Hadith is the best. Please refer to Figure 5 for the results. We followed the same steps for the MMR algorithm. You can check the similarity search and MMR results. Please refer to Figures 3 and 4.

The second part of the comparison is to compare the winning set of Hadith from the previous section to a Cohere reranker model search results.

4.5.4 Results Interpretation

Similarity Results Interpretation

Since all libraries returned the same inputs, there is nothing to compare here other than looking at the execution time. The results show that FAISS was the fastest, scoring an execution time of 1.35 seconds. This is understandable since it's free and the library files are stored locally on the server. The second one was Chroma, scoring an execution time of 2.89 seconds, and the slowest was Qdrant, scoring an execution time of 3.62 seconds. Refer to Figure 5 for details.

MMR Results Interpretation

As shown from the results, FAISS and Qdrant returned the same results, while Chroma changed the results slightly. Here, we asked the expert to tell us which set of Hadith will be considered more correct as a whole since both of them returned the same first value.

Refer to Figures 4 and 3 for details.

4.5.5 Expert Opinion

Sheikh Omar, our Imam at the mosque, is a distinguished scholar in Hadith. He has extensive knowledge of Hadith Science and the book *Riyad as-Salihin*.

Analysis of Search Results

We presented him with the results of searching for the word “money” in Arabic. The results were grouped into sets based on outputs from Chroma, FAISS, and similarity/re-ranking algorithms. These were provided for analysis and discussion (Google Docs). The discussion and related figures can be found in the discussion section.

Sheikh's Interpretation

The second group is structured in a logical order. It begins with a discussion on money, followed by its significance in people's lives, its distribution, and finally, guidelines on spending it.

In contrast, the first group lacks logical sequencing.

For example, when a father passes away, inheritance cannot be spent before it is properly divided. Therefore, understanding the principles of distribution should precede learning how to spend money.

From a devotional perspective, the second group is superior. It first establishes the significance of money for an individual before organizing the discussion in a structured and meaningful manner.

4.6 App / API Development

As the Hadith holds sacred significance for Muslims, we have decided to connect our Hadith API to a mobile app, making it available for everyone to use.

The app was developed using Flutter, a cross-platform technology that provides both Android and iOS compatibility. We initially focused on developing the Android version.

The app enables users to browse all the Hadith by navigating through each chapter. Additionally, it allows them to share and add Hadith to their favorites.

Furthermore, a dedicated search page is included, enabling users to conduct semantic searches across all Hadith. Users can also specify the number of returned Hadith.

Lastly, we have created a free API that provides access to individual Hadith, along with the capability to load more Hadith. This is embedded into the API to ensure better latency and reduce server load.

Links to both the API and the app are provided below.

- App link: https://bit.ly/waAhadeethAI
- API access: Please contact the first author via email to obtain access.

4.7 Conclusion

In this research, we applied Semantic Search and generative AI to the Hadith from the famous book Riyadh al-Salihin. First, we explained how we utilized ChatGPT 3 and 4 API to generate titles for the Hadith. We presented a comparison between the two language models (LLMs) and decided to proceed with the results from GPT-4. Additionally, we detailed our use of ChatGPT 3 for translating the titles from Arabic to English.

Subsequently, we described our application of Semantic Search to all the Hadith. We provided a comparison between different vector databases such as FAISS, Qdrant, and Chroma. Furthermore, we compared the results returned from search algorithms including similarity search, MMR, and cogere reranker.

Lastly, we offered expert opinions to assist in interpreting the results of Semantic Search across all the algorithms.

Finally, we explained how we created an app, API, dataset, and made them publicly available for download.

.1 Appendix A

.2 Figures

idInBook	matin_ar	matin_en	titles_ar_gpt3	title_ar_gpt4	title_en (Translated)
67	...قال: قال رسول الله صلى الله عليه وسلم : " من حسن إسلام المرء تركه ما لا يعنيه" ((حديث حسن رواه الترمذي وغيره))	Messenger of Allah (ﷺ) said, "It is from the excellence of (a believer's) Islam that he should shun that which is of no concern to him"..	حسن الإسلام	تجنب اللامهم في الإسلام	Avoiding Negligence in Islam
71	...أن النبي صلى الله عليه وسلم كان يقول: اللهم إني أسألك الهدى والتقى والعفاف والغنى" ((رواه مسلم)).	Allahumma inni as'alukal-huda wat- tuqa wal-'afafa wal-ghina (O Allah! I ask You for guidance, piety, chastity and self-sufficiency)"..	تقوى الدنيا والنساء	دعاء لطلب الهدى والتقوى والعفاف والغنى	Prayer for Guidance, Piety, Chastity, and Wealth
63	...قال: "إنكم لتعملون أعمالاً هي أدق في أعينكم من الشعر، كنا نعدها على عهد رسول الله صلى الله عليه وسلم من الموبقات" ((رواه البخاري)) 'وقال : الموبقات المهلكات '.	You indulge in (bad) actions which are more insignificant to you than a hair while we considered them at the time of Messenger of Allah (ﷺ) to be great destroying sins"..	الأعمال الدقيقة	التحذير من الاستهانة بالذنوب	Beware of Underestimating Sins
88	...قال: صليت وراء النبي صلى الله عليه وسلم بالمدينة العصر، فسلم ثم قام مسرعاً فتخطى رقاب الناس إلى بعض حجر نسائه، ففزع الناس من سرعته، فخرج عليهم، فرأى أنهم قد عجبوا من سرعته، قال "ذكرت شيئاً من تبر عندنا فكرهت أن يحبسنى، فأمرت بقسمته" ((رواه البخاري)). وفي رواية للبخاري : كنت خلفت في البيت تبراً من الصدقة، فكرهت أن أبيته ((التبر)) قطع ذهب أو فضة .	Once I performed the 'Asr prayer in Al-Madinah behind the Prophet (ﷺ). He (ﷺ) got up quickly after finishing the prayer with Taslim, and stepping over the people, went to one of the rooms of his wives. The people were startled at his haste, and when he came out and saw their astonishment at his urgency he said, "I recalled that there was left with me some gold which was meant for charity; I did not like to keep it any longer, so I gave orders that it should be distributed"..In another narration, Messenger of Allah said, "I had left some gold for Sadaqah in the house, and did not wish to keep it overnight".	فضل التبرع بالسرعة	عجلة النبي صلى الله عليه وسلم لقسمة التبر	The Prophet's Urgency in Distributing Charity
1521	قال: قال لي رسول الله صلى الله عليه وسلم : "لا تحقرن من المعروف شيئاً، ولو أن تلقي أخاك بوجه طلق" ((رواه مسلم)).	The Messenger of Allah (ﷺ) said to me, "Do not belittle any good deed, even your meeting with your brother (Muslim) with a cheerful face.".	قيمة المعروف	عظمة الأعمال الصغيرة في الإسلام	The Greatness of Small Businesses in Islam

Figure 2: Gpt3.5 vs Gpt 4 Generated titles

System Prompts	Purpose	Model
Generate a title that can best describe the Hadeeth, be concise. Hadeeth: {hadeeth}	Generating Arabic titles to Hadith	gpt-4-turbo
Translate given title to English. Return only the translated title. Title: {title}	Translating Arabic titles to English titles	gpt-3.5-turbo-0125

Table 1: System prompts used in research

Query = "المال" n=5	Chroma mmr (English)	Faiss &. Qdrant mmr (same results is English)
Resutls	**The Nation's Crisis: Money** Messenger of Allah (ﷺ) said, "Verily, there is a Fitnah (trial) for every nation and the trial for my nation (or Ummah) is wealth." **Spending in the Way of Allah from Wealth One Loves** Abu Talhah (May Allah be pleased with him) was the richest among the Ansar of Al-Madinah and possessed the largest property; and among his possessions what he loved most was his garden known as Bairuha' which was opposite the mosque, and Messenger of Allah (ﷺ) often visited it and drank from its fresh water. When this ayah was revealed: "By no means shall you attain Al-Birr (piety, righteousness - here it means Allah's reward, i.e., Jannah), unless you spend (in Allah's Cause) of that which you love," (3:92). Abu Talhah came to Messenger of Allah (ﷺ), and said: "Allah says in His Book: 'By no means shall you attain Al-Birr, unless you spend (in Allah's Cause) of that which you love,' and the dearest of my property is Bairuha' so I have given it as Sadaqah (charity) for Allah's sake, and I anticipate its reward with Him; so spend it, O Messenger of Allah, as Allah guides you". Messenger of Allah (ﷺ) said, "Well-done! That is profitable property. I have heard what you have said, but I think you should spend it on your nearest relatives." So Abu Talhah distributed it among his nearest relatives and cousins. **Words for Sufficiency and Wealth** A slave who had made a contract with his master to pay for his freedom, came to me and said: "I am unable to fulfill my obligation, so help me." He said to him: "Shall I not teach you a supplication which the Messenger of Allah (ﷺ) taught me? It will surely prove so effective that if you have a debt as large as a huge mountain, Allah will surely pay it for you. Say: 'Allahumm-akfini bihalalika 'an haramika, wa aghnini bifadlika 'amman siwaka (O Allah! Grant me enough of what You make lawful so that I may dispense with what You make unlawful, and enable me by Your Grace to dispense with all but You).'" **Real Estate Trust and Just Governance** The Prophet (ﷺ) said, "A man bought a piece of land from another man, and the buyer found a jar filled with gold in the land. The buyer said to the seller: 'Take your gold, as I bought only the land from you and not the gold.' The owner of the land said: 'I sold you the land with everything in it.' So both of them took their case before a third man who asked: 'Have you any children?' One of them said: 'I have a boy.' The other said, 'I have a girl.' The man said: 'Marry the girl to the boy and spend the money on them; and whatever remains give it in charity.'" **Real Wealth and Inheritance** Messenger of Allah (ﷺ) asked, "Who of you loves the wealth of his heir more than his own wealth?" The Companions said: "O Messenger of Allah! There is none of us but loves his own wealth more." He (ﷺ) said, "His wealth is that which he has sent forward, but that which he retains belongs to his heir."	**The Nation's Crisis: Money** Messenger of Allah (ﷺ) said, "Verily, there is a Fitnah (trial) for every nation and the trial for my nation (or Ummah) is wealth" (Reported by At-Tirmidhi and he said: Hadith Hasan Sahih). **What Remains with the Dead** The Prophet (ﷺ) said, "Three follow a dead body: members of his family, his possessions and his deeds. Two of them return and one remains with him. His family and his possessions return; his deeds remain with him" (Agreed upon). **Real Estate Trust and Just Governance** The Prophet (ﷺ) said, "A man bought a piece of land from another man, and the buyer found a jar filled with gold in the land. The buyer said to the seller: 'Take your gold, as I bought only the land from you and not the gold.' The owner of the land said: 'I sold you the land with everything in it.' So both of them took their case before a third man who asked: 'Have you any children?' One of them said: 'I have a boy.' The other said, 'I have a girl.' The man said: 'Marry the girl to the boy and spend the money on them; and whatever remains give it in charity'" (Agreed upon). **Spending in the Way of Allah from Wealth One Loves** Abu Talhah (May Allah be pleased with him) was the richest among the Ansar of Al-Madinah and possessed the largest property; and among his possessions what he loved most was his garden known as Bairuha' which was opposite the mosque, and Messenger of Allah (ﷺ) often visited it and drank from its fresh water. When this ayah was revealed: "By no means shall you attain Al-Birr (piety, righteousness - here it means Allah's reward, i.e., Jannah), unless you spend (in Allah's Cause) of that which you love," (3:92). Abu Talhah came to Messenger of Allah (ﷺ), and said: "Allah says in His Book: 'By no means shall you attain Al-Birr, unless you spend (in Allah's Cause) of that which you love,' and the dearest of my property is Bairuha' so I have given it as Sadaqah (charity) for Allah's sake, and I anticipate its reward with Him; so spend it, O Messenger of Allah, as Allah guides you." Messenger of Allah (ﷺ) said, "Well-done! That is profitable property. I have heard what you have said, but I think you should spend it on your nearest relatives." So Abu Talhah distributed it among his nearest relatives and cousins (Agreed upon). **Words for Sufficiency and Wealth** A slave who had made a contract with his master to pay for his freedom, came to me and said: "I am unable to fulfill my obligation, so help me." He said to him: "Shall I not teach you a supplication which the Messenger of Allah (ﷺ) taught me? It will surely prove so effective that if you have a debt as large as a huge mountain, Allah will surely pay it for you. Say: 'Allahumm-akfini bihalalika 'an haramika, wa aghnini bifadlika 'amman siwaka (O Allah! Grant me enough of what You make lawful so that I may dispense with what You make unlawful, and enable me by Your Grace to dispense with all but You)" (Reported by At-Tirmidhi and he said: Hadith Hasan).

Figure 3: MMR search Results (Arabic)

Libraries: Chroma, FAISS and Qdrant Results (all of them returned the same results) **Algorthem:** similarity search **Query =** "المال" n=5		
Results	**فتنة الأمة: المال** قال سمعت رسول الله، صلى الله عليه وسلم ، يقول: "" إن لكل أمة فتنة، وفتنة أمتى المال"" ((رواه الترمذي وقال: حديث حسن صحيح)). **الغنى الحقيقي هو غنى النفس** قال: ""ليس الغنى عن كثرة العرض، ولكن الغنى غنى النفس"" ((متفق عليه)). **حقيقة المال والانشغال به** قال: أتيت النبي ، صلى الله عليه وسلم، وهو يقرأ: { ألهاكم التكاثر} قال: "" يقول ابن آدم: مالي، وهل لك يا ابن آدم من مالك إلا ما أكلت فأفنيت، أو لبست فأبليت، أو تصدقت فأمضيت؟ !"" ((رواه مسلم)). **الإنفاق في سبيل الله** قال: ""لو كان لي مثل أحد ذهباً، لسرني أن لا تمر على ثلاث ليال وعندي منه شئ إلا شئ أرصده لدين"" ((متفق عليه)) **المال الحقيقي والميراث** قال: قال رسول الله صلى الله عليه وسلم : "أيكم مال وارثه أحب إليه من ماله؟" قالوا: يا رسول الله، ما منا أحد إلا ماله أحب إليه. قال: ""فإن ماله ما قدم ومال وارثه ما أخر"" ((رواه البخاري)).	**The Nation's Crisis: Money** Messenger of Allah (ﷺ) said, ""Verily, there is a Fitnah (trial) for every nation and the trial for my nation (or Ummah) is wealth."" **The true wealth is the wealth of the soul** The Prophet (ﷺ) said, ""Richness is not the abundance of wealth, rather it is self-sufficiency."". **The Truth About Money and Being Preoccupied with It** I came to the Prophet (ﷺ) while he was reciting (Surat At-Takathur 102):""The mutual rivalry (for hoarding worldly things) preoccupy you. Until you visit the graves (i.e., till you die). Nay! You shall come to know! Again nay! You shall come to know! Nay! If you knew with a sure knowledge (the end result of hoarding, you would not have been occupied in worldly things). Verily, you shall see the blazing Fire (Hell)! And again, you shall see it with certainty of sight! Then (on that Day) you shall be asked about the delights (you indulged in, in this world)!"" (102:1-8)(After reciting) he (ﷺ) said, ""Son of adam says: 'My wealth, my wealth.' Do you own of your wealth other than what you eat and consume, and what you wear and wear out, or what you give in Sadaqah (charity) (to those who deserve it), and that what you will have in stock for yourself."". **Spending in the Way of Allah** Messenger of Allah (ﷺ) said: ""If I had gold equal to Mount Uhud (in weight), it would not please me to pass three nights and I have a thing of it left with me, except what I retain for repayment of a debt"".. **Real Wealth and Inheritance** Messenger of Allah (ﷺ) asked, ""Who of you loves the wealth of his heir more than his own wealth?"" The Companions said: ""O Messenger of Allah! There is none of us but loves his own wealth more."" He (ﷺ) said, ""His wealth is that which he has sent forward, but that which he retains belongs to his heir.""
Time execuation for Chroma (n=5)	Execution time: 2.89 seconds	
Time execuation for FAISS (n=5)	**Execution time: 1.35 seconds (Fastest)**	
Time execuation for Qdrant (n=5)	Execution time: 3.62 seconds	

Figure 4: MMR search Results (English)

Libraries: Chroma, FAISS and Qdrant Results (all of them returned the same results) **Algorthem:** similarity search **Query =** "المال" **n**=5		
Results	**فتنة الأمة: المال** قال سمعت رسول الله، صلى الله عليه وسلم ، يقول: "" إن لكل أمة فتنة، وفتنة أمتى المال"" ((رواه الترمذي وقال: حديث حسن صحيح)). **الغنى الحقيقي هو غنى النفس** قال: ""ليس الغنى عن كثرة العرض، ولكن الغنى غنى النفس"" ((متفق عليه)). **حقيقة المال والانشغال به** قال: أتيت النبي ، صلى الله عليه وسلم، وهو يقرأ: { ألهاكم التكاثر} قال: "" يقول ابن آدم: مالي، وهل لك يا ابن آدم من مالك إلا ما أكلت فأفنيت، أو لبست فأبليت، أو تصدقت فأمضيت؟ !"" ((رواه مسلم)). **الإنفاق في سبيل الله** قال: ""لو كان لي مثل أحد ذهباً، لسرني أن لا تمر على ثلاث ليال وعندي منه شئ إلا شئ أرصده لدين"" ((متفق عليه)) **المال الحقيقي والميراث** قال: قال رسول الله صلى الله عليه وسلم : "أيكم مال وارثه أحب إليه من ماله؟" قالوا: يا رسول الله، ما منا أحد إلا ماله أحب إليه. قال: ""فإن ماله ما قدم ومال وارثه ما أخر"" ((رواه البخاري)).	**The Nation's Crisis: Money** Messenger of Allah (ﷺ) said, ""Verily, there is a Fitnah (trial) for every nation and the trial for my nation (or Ummah) is wealth."" **The true wealth is the wealth of the soul** The Prophet (ﷺ) said, ""Richness is not the abundance of wealth, rather it is self-sufficiency."". **The Truth About Money and Being Preoccupied with It** I came to the Prophet (ﷺ) while he was reciting (Surat At-Takathur 102):""The mutual rivalry (for hoarding worldly things) preoccupy you. Until you visit the graves (i.e., till you die). Nay! You shall come to know! Again nay! You shall come to know! Nay! If you knew with a sure knowledge (the end result of hoarding, you would not have been occupied in worldly things). Verily, you shall see the blazing Fire (Hell)! And again, you shall see it with certainty of sight! Then (on that Day) you shall be asked about the delights (you indulged in, in this world)!"" (102:1-8)(After reciting) he (ﷺ) said, ""Son of adam says: 'My wealth, my wealth.' Do you own of your wealth other than what you eat and consume, and what you wear and wear out, or what you give in Sadaqah (charity) (to those who deserve it), and that what you will have in stock for yourself."". **Spending in the Way of Allah** Messenger of Allah (ﷺ) said: ""If I had gold equal to Mount Uhud (in weight), it would not please me to pass three nights and I have a thing of it left with me, except what I retain for repayment of a debt"".. **Real Wealth and Inheritance** Messenger of Allah (ﷺ) asked, ""Who of you loves the wealth of his heir more than his own wealth?"" The Companions said: ""O Messenger of Allah! There is none of us but loves his own wealth more."" He (ﷺ) said, ""His wealth is that which he has sent forward, but that which he retains belongs to his heir.""
Time execuation for Chroma (n=5)	Execution time: 2.89 seconds	
Time execuation for FAISS (n=5)	**Execution time: 1.35 seconds (Fastest)**	
Time execuation for Qdrant (n=5)	Execution time: 3.62 seconds	

Figure 5: similarity search Results (Arabic)

I will give you a Text. It is a Hadith from Prophet Mohamed. we need to generate a title that can describe the Hadeeth
return results as json object, key is title.
Example 1:
Hadeeth:
إنما الأعمال بالنيات، وإنما لكل امرىء ما نوى فمن كانت هجرته إلى الله ورسوله فهجرته إلى الله ورسوله، ومن كانت هجرته لدنيا يصيبها، أو امرأة ينكحها فهجرته إلى ما هاجر إليه
title: نية الأعمال

Example 2:
Hadeeth:
"صلاة الرجل في جماعة تزيد على صلاته في سوقه وبيته بضعاً وعشرين درجة وذلك أن أحدهم إذا توضأ فأحسن الوضوء ثم أتى المسجد لا يريد إلا الصلاة، لا ينهزه إلا الصلاة، لم يخط خطوة إلا رفع له بها درجة، وحط عنه بها خطيئة حتى يدخل المسجد، فإذا دخل المسجد كان في الصلاة ما كانت الصلاة هي تحبسه، والملائكة يصلون على أحدكم ما دام في مجلسه الذي صلى فيه، مالم يحدث فيه"
((متفق عليه، وهذا لفظ مسلم)).
وقوله صلى الله عليه وسلم: ' ينهزه هو بفتح الياء والهاء بالزاي: أى يخرجه وينهضه
title: فضل صلاة الجماعة

Example 3:
Hadeeth:
"لا هجرة بعد الفتح، ولكن جهاد ونية، وإذا استنفرتم فانفروا"
title:
"الجهاد والنية بعد الفتح"

Example 4:
سئل رسول الله صلى الله عليه وسلم عن الرجل يقاتل شجاعة، ويقاتل حميةً، ويقاتل رياء، أي ذلك في سبيل الله؟ فقال رسول الله صلى الله عليه وسلم:
" من قاتل لتكون كلمة الله هى العليا فهو في سبيل الله"
title :
"القتال في سبيل الله"

Figure 6: Prompts used for title generation using GPT3

.3 System Prompts, Purpose, Model

Chapter Transition: The successful deployment of the Ahadeeth AI pipeline established a fundamental ”ground truth” requirement for the research moving forward. It verified that LLMs could indeed parse and retrieve complex classical Arabic literature accurately, provided the data was rigidly structured (Matn vs. Sanad) and retrieved under strict mathematical controls (Semantic Search and Reranking) rather than open generation. However, retrieving pre-existing text is only half the battle. In the context of actual Islamic Jurisprudence, individuals ask direct, novel questions that require legally binding answers (Fatawa). Moving from mere retrieval to actively answering complex theological questions via AI required the creation of an entirely new Agentic RAG architecture. This architecture needed to guarantee zero-hallucination outputs while dynamically cross-referencing massive volumes of Hanafi Jurisprudence. The following chapters explore this ultimate challenge, detailing the systemic framework of Fatawa AI and the flagship Agentix Islam RAG framework.

Chapter A

Fatawa AI: Identifying the Hallucination Deficit in LLMs

A.1 Introduction

The rapid proliferation and widespread application of Artificial Intelligence, particularly Large Language Models (LLMs), has begun to intersect with Islamic Studies in profound ways, presenting an unprecedented blend of opportunities and ethical challenges [64]. While AI has been previously and successfully explored in the context of Hadith authentication, preservation, and modernized semantic search through projects like Ahadeeth.AI, a much more complex frontier lies ahead: the computational generation of religious rulings (Fatawa) according to specific schools of thought.

The domain of Fatawa is fundamentally different from general knowledge retrieval. Issuing a Fatwa requires deep contextual awareness, an understanding of the questioner's exact circumstances, and the ability to navigate centuries of complex analogical reasoning (*qiyas*), juristic preference (*istihsan*), and local customs (*urf*). This is particularly true for the Hanafi school of jurisprudence, which is historically renowned for its intricate use of reason, detailed hypothetical scenario planning, and highly contextual legal derivations.

To quantify the capabilities and deficiencies of LLMs in navigating this complex domain, new benchmarks have progressively emerged, such as IslamicLegalBench [65] and FiqhQA [66]. These benchmarks attempt to evaluate LLMs on a spectrum of Islamic jurisprudence knowledge, moving beyond simple factual recall into the realm of legal reasoning.

Building upon these foundational efforts, our study shifts focus to a highly targeted evaluation. We investigate whether modern, commercially available LLMs, when acting strictly bounded by their pre-trained weights without external grounding, can be trusted to provide reliable and accurate Fatawa specifically within the Hanafi school. Unlike FiqhQA and IslamicLegalBench which evaluate broad, multi-school capabilities and primarily fact-based QA pairs, our approach introduces a massive, strictly restricted deep-dive constraint into a single Fiqh school, demanding precise reasoning rather than generic recall. We demonstrate that despite their general capabilities, these models struggle significantly with the precision demanded by Islamic law.

A.2 Context and Prior Work

A.2.1 Ahadeeth.AI and Islamic Information Retrieval

The preservation and accessible searchability of foundational Islamic texts has always been a scholarly priority. Riyadh al-Salihin, for example, is a cornerstone of Hadith literature comprising 1,896 authenticated narrations that dictate ethical and practical Muslim life. Traditionally, searching this corpus, and similar bodies of work, relied entirely on exact lexical matching. This rigid approach often misses the underlying linguistic intent or the broader theological context, especially when dealing with classical Arabic syntax which is highly inflectional and rich in synonyms.

By leveraging LLMs and the power of semantic search, we allow users to search the corpus by meaning, concept, and thematic relevance rather than strict keyword alignment. Our prior work involved utilizing Generative AI (GPT-4) to systematically create highly descriptive, modern titles and summaries for ancient texts. We then applied OpenAI's text-embedding-3-large model, combined with high-performance Vector Databases such as FAISS, Chroma, and Qdrant, alongside Cohere's rerank-multilingual-v2.0. This modern pipeline significantly modernized accessibility, demonstrating that while AI is incredibly potent as an organizing and retrieval tool for static, verified texts, generating novel rulings from scratch is an entirely different challenge.

A.2.2 LLMs in Fatawa Generation

The idea of utilizing Large Language Models to autonomously generate Fatawa has sparked significant debate across the Islamic and technological world. Owing to the nature of their pre-training data which often underrepresents classical, nuanced Arabic in favor of modern standard Arabic or translated texts, LLMs exhibit a dangerous propensity to "hallucinate". In a religious context, a hallucination is not merely an error; it can manifest as a fabricated Arabic citation attributed to a canonical scholar, or the dangerous conflation of distinct, sometimes opposing, schools of jurisprudence.

To address these severe shortcomings systematically, the academic NLP community has established targeted initiatives. For instance, the IslamicEval 2025 shared task was explicitly aimed at detecting, penalizing, and mitigating hallucinations in Islamic text generation [64]. Parallelly, massive multi-school benchmarks such as IslamicLegalBench [65] and FiqhQA [66] have emerged to quantify these failures. IslamicLegalBench, drawing from 1,200 years of pluralistic legal traditions, starkly revealed that state-of-the-art LLMs exhibit significant limitations, hallucinating frequently and failing to achieve high accuracy on complex legal reasoning tasks. This underscores a clear imperative for rigorous, school-specific evaluations, mitigating the “blur” between different Fiqh methodologies, and deeply motivating our targeted dive into Hanafi Fiqh explored in this paper.

A.3 Fatawa AI: Dataset and Methodology

We rigorously focus our evaluation strictly on Islamic Jurisprudence (Fiqh) rulings from the Hanafi school of thought. Evaluating a model on a single, defined school prevents the model from being penalized for providing a correct answer according to the Shafi’i or Maliki school when a Hanafi answer was expected, thereby isolating its capacity to recall and reason within one specific boundary condition.

A.3.1 Dataset Construction and Validation

Our dataset is a highly structured, proprietary compilation consisting of 10,000 discrete Questions and Answers (Q&A) pairs. These pairs exclusively represent rulings curated from the Hanafi school. Because web-scraped data is notoriously noisy and unreliable in religious contexts, every answer in our dataset was expert-validated through iterative review; final adjudication was performed by a recognized human domain expert.

Initially, we scraped and collected raw Q&A data into an unstructured JSON format from verified digital Islamic archives. The human domain expert then designed a comprehensive thematic structure, categorizing the sprawling domain of Fiqh into 10 main pillars and 92 highly specific sub-categories. This taxonomy ensures that our dataset covers a representative distribution of daily life scenarios—ranging from modern financial transactions to classical purification rituals.

To map our 10,000 Q&A pairs into this taxonomy at scale, we employed Cosine Similarity on Q&A text embeddings. Each question and its corresponding answer were embedded and matched against the vector representation of the 92 sub-categories. Following this automated sorting, the domain expert performed an exhaustive manual review and validation of the categorizational outputs. Through several iterations of error-correction, we established high categorical consensus and verified theological accuracy based on expert evaluation. This manual effort guarantees that our evaluation baseline remains extremely robust for targeted inference.

A.3.2 Evaluation Methodology: The LLM-as-a-Judge Paradigm

Our primary experimental goal is to determine if baseline, commercially available LLMs can accurately answer Hanafi School questions relying solely on their internal, pre-trained parametric knowledge, without the aid of external document retrieval (RAG). To maintain computational efficiency while ensuring statistical significance, we extracted a perfectly balanced sample of 300 questions from the dataset using uniform stratified random sampling with *random_seed=42*. These 300 questions were distributed evenly across the top 5 most popular overarching categories (exactly 60 questions dedicated to each category).

The models selected for this evaluation are Gemini 2.5 Flash (*gemini-2.5-flash*) via Google GenAI Studio, and GPT-5.1 (*gpt-5.1-0125*) via OpenAI API. The evaluation snapshot was taken on March 1, 2026, representing the current frontier of accessible, highly optimized large language models. For precise reproducibility, GPT-5.1 was constrained with *reasoning="low"*, and Gemini 2.5 Flash was executed with *thinking_budget=-1*. The prompt provided to each model strictly instructed it to act as a Hanafi scholar and provide a concise, direct ruling to the user's question (see Appendix A).

Given the linguistic complexity of Fiqh and the fact that a correct ruling can be phrased in dozens of different valid ways, simple string matching (like Exact Match or BLEU scores) is entirely inadequate for grading. Therefore, we employed Anthropic's Claude 3.5 Sonnet (*claude-3-5-sonnet-20241022*), initialized with *thinking="enabled"* and *max_tokens=10000*, acting as an advanced "LLM-as-a-judge". Claude 3.5 Sonnet was tasked with comparing the generated answer from Gemini or GPT against the human domain expert's verified ground-truth answer. The judge was instructed to look beyond syntax and strictly evaluate the semantic, legal equivalence of the ruling.

To provide a granular understanding of model failure, responses were judged and scored on a rigid three-point scale:

- **-1 (Contradictory)**: The model provided an answer that is fundamentally opposite to the ground truth (e.g., stating an act is 'Haram' (forbidden) when the Hanafi ruling is 'Halal' (permissible), or applying the wrong mathematical formula in inheritance). This is the most dangerous failure mode.

- **0 (Partially Correct)**: The model agrees with the ground truth in some fundamental aspects but hallucinates conditions, misses critical exceptions, or blends Hanafi rulings with those of another school. While less dangerous, it demonstrates a lack of authoritative clarity.

- **1 (Identical Meaning)**: The generated answer is semantically identical in intent, condition, and ruling to the domain expert's ground truth, representing a safely deployable response.

A.4 Results and Analytical Discussion

We evaluate the models primarily through the lens of Simple Accuracy. Simple Accuracy in this context is strictly defined as the percentage of output results that achieved a perfect **+1** score (Identical meaning) from the LLM judge. We purposefully do not grant partial credit in the overarching accuracy metric, as a "partially correct" Fiqh ruling often lacks the necessary precision to be safely acted upon by a layman.

$$Accuracy = \frac{\text{Count of +1 Scores}}{\text{Total Number of Results}} \times 100\% \quad \text{(A.1)}$$

A.4.1 Overall Baseline Accuracy

Table A.1 presents the aggregate performance of the evaluated models across the entirety of the strictly balanced 300-question subset. Gemini-2.5-Flash achieved an overall accuracy of 46.33% (139/300) with a 95% Confidence Interval of [40.7%, 52.0%], subsequently outperforming GPT-5.1, which secured 43.67% (131/300) with a 95% Confidence Interval of [38.0%, 49.3%], by a margin of 2.66 percentage points. To evaluate the statistical significance of this difference, we applied McNemar's test for paired nominal data; the resulting statistic ($\chi^2 = 0.54$, $p = 0.46$) indicates no statistically significant difference in performance between the two models.

This baseline metric is highly illuminating. It suggests that despite containing trillions of parameters and demonstrating remarkable fluency in general Arabic, both models fall drastically short of a reliable threshold (often considered to be $> 95\%$ in medical or legal AI applications) when dealing with specialized jurisprudential knowledge. In over 53% of all cases, the models either provided a contradictory ruling, or a ruling polluted with unverified conditions. Specifically, examining the global class distribution of the judge labels, Gemini yielded 94 Contradictory (-1) outputs and 67 Partially Correct (0) outputs, while GPT-5.1 yielded 111 Contradictory (-1) outputs and 58 Partially Correct (0) outputs. The slight but notable difference between Gemini and GPT-5.1 points to variations in their pre-training mixture regarding classical text exposure, but underscores a universal architecture limitation in reasoning without grounded retrieval.

A.4.2 Cross-Category Performance and Logic Breakdown

To better understand the strengths and weaknesses of these models, we performed a deep-dive analysis of performance across the 5 distinct categories tested (see Table A.2).

LLM Model	Evaluated (N)	Correct (+1)	Accuracy
GPT-5.1	300	131	43.67%
Gemini-2.5-Flash	300	139	46.33%

Table A.1: Total Evaluation and Simple Accuracy across all 300 tested Hanafi Fiqh instances.

The nature of the context overwhelmingly dictates the model's reliability. The semantic and conceptual boundaries between categories highlight that Fiqh is not a monolithic database of facts, but a diverse landscape of required reasoning skills.

Gemini-2.5-Flash demonstrated a marked and surprising superiority in handling rulings related to *Dhikr* (Ethics/Purification) and *Ibadat* (Acts of Worship). These two categories generally rely on well-documented, highly repetitive classical texts detailing daily rituals (e.g., how to perform Wudu, conditions of prayer). Gemini's higher accuracy here likely stems from a broader internalization of these highly prominent, frequently repeated ritualistic texts during its pre-training phase.

Conversely, GPT-5.1 demonstrated a much stronger grasp of *Zawaj* (Family Law, Marriage, Divorce) and *Hazr* (Prohibition, Dress, Adornment). These categories shift away from simple repetitive rituals and introduce complex, conditional human scenarios. Family Law, in particular, involves multi-step logical deductions (e.g., "If condition A occurs and condition B is absent, then the divorce is revocable"). GPT-5.1's superior architecture in handling chained logic and complex zero-shot reasoning likely contributes to its higher peak accuracy (55.00%) in the Zawaj category.

Performance in *Mu'amalat* (Transactions, Commerce, Trusts) was universally poor across both models (ranging roughly between 41% and 42%). This is arguably the most mathematically and logically dense branch of Fiqh, governing modern financial analogues against classical contract law. The models consistently failed to apply ancient transactional conditions to modernized hypotheticals, frequently hallucinating modern economic principles in place of strict Hanafi contract stipulations. Let this serve as a clear indicator that without external grounding, LLMs cannot currently navigate the complexities of Islamic finance reliably.

Category (Arabic)	Category (English)	GPT-5.1 Accuracy	Gemini-2.5-Flash Accuracy	Winner
Mu'amalat	Transactions and Trusts	41.67% (25/60)	41.67% (25/60)	Tie
Hazr	Prohibition, Dress, Adornment	46.67% (28/60)	43.33% (26/60)	GPT-5.1
Dhikr	Remembrance, Purification, Ethics	40.00% (24/60)	**51.67%** (31/60)	Gemini-2.5-Flash
Zawaj	Marriage, Divorce, Expenditures	**55.00%** (33/60)	51.67% (31/60)	GPT-5.1
Ibadat	Acts of Worship	35.00% (21/60)	43.33% (26/60)	Gemini-2.5-Flash

Table A.2: Detailed Cross-Category Performance Comparison illustrating model strength variations across fundamentally different logic domains.

A.5 Discussion and Ethical Implications

The data generated from this evaluation carries profound implications for the deployment of AI in religious contexts. An overarching accuracy hovering in the mid-40th percentile confirms that querying a base LLM for Islamic legal advice is currently an unsafe practice. The fundamental flaw observed is not just the lack of knowledge, but the confident presentation of incorrect knowledge.

When an LLM scores a 0 (Partially Correct) or a -1 (Contradictory), it rarely signals uncertainty to the user. Instead, it generates an authoritative-sounding Fatwa, often cloaked in classical Arabic formatting, which mimics the structure of an authentic scholarly ruling. For an average layman, distinguishing between a deeply grounded Hanafi ruling and a beautifully articulated hallucination is practically impossible. This phenomenon directly supports the urgent warnings from the traditional scholarly community. Prominent Muftis and Islamic scholars consistently warn against the intrinsic dangers of utilizing AI as a standalone, primary source for religious rulings [67]. The core of this concern lies in the realization that AI systems inherently lack moral accountability. They operate in a vacuum devoid of contextual awareness—unable to read the emotional distress of a questioner, incapable of evaluating local traditions, and completely lacking the independent, grounded reasoning capacity (*ijtihad*) necessary to interpret rulings in alignment with the higher objectives of Islamic law (Maqasid al-Shari'ah).

The deployment of ungrounded LLMs in this space risks widespread dissemination of religious inaccuracies, which can severely impact individual observances and family law implementations. Therefore, we posit that the "Zero-Shot" or "Direct Query" method for religious rulings should be actively discouraged by developers building Islamic LLM tooling. Instead, deployments must enforce a strict 'Human-in-the-Loop' (HITL) policy—where the AI serves exclusively as a retrieval and drafting understudy, requiring mandatory review and final authorization by a credentialed Mufti before any output reaches the end-user.

A.6 Conclusion and Future Work

Our findings empirically demonstrate that while Large Language Models possess fascinating capabilities and the raw potential to understand and process Jurisprudence questions, they currently struggle massively to fulfill the requirements of expert-level reliability. With peak accuracies hovering around only 55% in the absolute best-case scenarios (GPT-5.1 navigating Family Law), base models cannot be trusted as standalone Muftis. Their internal weights represent an amalgamation of conflicting data rather than a strict, organized knowledge hierarchy of a single Fiqh school.

A primary limitation of our study is the reliance on Claude 3.5 Sonnet as an automated judge. While LLM-as-a-judge is a recognized paradigm within NLP evaluation, future work must establish rigorous human-inter-rater reliability against Claude's scoring within the highly sensitive domain of Fiqh to fully validate the evaluation metric itself. Specifically, a human expert must re-annotate a random subset of the model outputs (e.g., $n = 100$) and report the agreement (using Cohen's Kappa or Krippendorff's Alpha) against Claude's assigned labels. Furthermore, expanding the evaluation corpus to include specifically fine-tuned Arabic LLMs (e.g., Jais or ALLaM) may reveal different architectural efficiencies.

Most importantly, our future roadmap includes comparing these baseline parametric models against advanced Retrieval-Augmented Generation (RAG) systems. Given the substantial failure rates in zero-shot reasoning, we hypothesize that limiting the LLM to strictly synthesize retrieved, pre-authenticated Hanafi texts (rather than generating from memory) is the only viable path to safety. Ultimately, an AI Agent designed exclusively for Fatwa generation based on a structured RAG approach shows promising potential. As demonstrated by early adoption tests where such systems successfully answered over 4,000 questions while acting solely as a safe research assistant to human domain experts, the future of Fatawa AI lies not in autonomous ruling generation, but in massive, intelligent retrieval scaling.

.1 Appendix: Evaluation Prompt Template

For exact reproducibility, the following system prompt was provided to both Gemini-2.5-Flash and GPT-5.1 during the generation phase. The prompt enforces the model to act as a strict Hanafi scholar and mitigates arbitrary stylistic verbosity:

> *"You are an expert Mufti and scholar in the Hanafi school of Islamic Jurisprudence (Fiqh). Your task is to provide a direct, concise, and accurate Fatwa to the user's question. Do not provide information from other schools of thought (like Shafi'i, Maliki, or Hanbali) unless explicitly asked. State the ruling clearly at the beginning."*

Chapter A

Agentix Islam: Architecting the Zero-Hallucination Framework

A.1 Technical Implementation of the Agentic Islamic Fatwa (RAG) Framework

This document details the exact methodology, architecture, and logic used to construct the custom Retrieval-Augmented Generation (RAG) system for the Islamic Fatwa agent (as seen in the 'R6-Agentix Islam (RAG)' project). This text is formulated to be easily integrated into the "Implementation" chapter of your PhD Thesis.

A.1.1 1. Data Ingestion Structured Parsing

The foundational step involved digitizing and parsing classical Islamic jurisprudence texts into a machine-readable format while strictly preserving scholarly integrity.

* **Batch Processing with LLMs:** The system uses the **Gemini 2.5 Pro** model via the Batch API to process raw manuscript pages at scale. * **JSON Schema Enforce-**

ment: To ensure structural consistency, a strict JSON schema was provided in the system prompt. The model was instructed to extract content page-by-page into a strict dictionary containing: * ‘page$_{c}ontent‘ : Themaintextofthepage. * ‘page_{n}umber‘ : Thespecific, verifiablepagenumbertoensureaccuratecitationlater. *‘notes‘ : Aspecializedsub-objectdesignedtouniquelycaptureclassicaltextannotations, heavilyseparating‘footnote‘s(Footnote$

A.1.2 2. Hierarchical Metadata Table of Contents (TOC) Engineering

An intelligent RAG system requires semantic understanding of the book’s structure, not just isolated flat text strings.

* **TOC Flattening:** The books’ structural Table of Contents were mapped into a hierarchical JSON array. Each section, chapter, or topic (e.g., ”Introduction to the New Edition”) was mapped with a ‘title‘, ‘startPage‘, ‘endPage‘, and its nesting ‘level‘ or ‘parentId‘ (pathIds and pathTitles).

* **Markdown Synthesis:** The extracted raw JSON pages were subsequently stitched together to form a clean, continuous Markdown structure. Crucially, page numbers and separated footnotes were interleaved at designated points, maintaining the text’s academic validity while making it parsable for chunking algorithms.

A.1.3 3. Vectorization and Database Construction

Before being queried, the structured text was transformed into embeddings designed for high-precision semantic search.

Intelligent Text Splitting: *The compiled Markdown documents were chunked using LangChain’s ‘RecursiveCharacterTextSplitter‘. Because Islamic jurisprudence requires high context retention, the chunks were set to ‘1000‘ characters with a safety overlap of ‘200‘ characters to prevent cutting off crucial conditions or* Ahadith*. * **Metadata Tag-**

ging: Every individual chunk was injected with localized metadata referencing its origin: ‘book$_{i}d‘, ‘book_{n}ame‘, ‘book_{p}art_{n}umber‘(volume), andtheexact‘page_{n}umber‘.*$

Embeddings Storage:$Textchunkswerevectorizedusing Google's‘gemini-embedding-$

$001‘modelandstoredimmutablyintoacloud-hosted$**Chroma DB**$(withanarchitecturalfallbacktolocal$

Two-Stage Retrieval Mechanism (MMR + Reranking):$Tosolvethepervasiveissueofirrelevantfatw$

1.**Initial Retrieval:**$MaximumMarginalRelevance(MMR)fetchesthetopk$=$5diversebuthighlyrelev$

$scoreandcompressthereturnednodes, pushingthemostsemanticallyrelevanttheologicalrulingstothe$

A.1.4 4. Agentic Orchestration and Backend Logic (FastAPI)

The central intelligence of the framework is a dynamic orchestrator built with FastAPI, deploying **Gemini 3 Flash Preview** (and Gemini 2.5 Flash). It functions autonomously as an ”Islamic Scholar” agent.

* **Strict RAG Guardrails:** The LLM is given a definitive system instruction commanding it to act strictly as an Islamic scholar. It is forced to answer **only** from the retrieved context. If the ruling does not exist, it must declare its inability to answer (”Information not available in the provided sources”) instead of hallucinating. It must answer in Arabic or English and maintain scholarly etiquette.

* **Function Calling (The Agentic Component):** Rather than blindly performing similarity searches on every query, the LLM is equipped with specialized tools (functions): * ‘query_vector_store‘: Triggered automatically for general theological queries across the whole database. * ‘find_page_scopes‘ & ‘get_pages_in_range‘: Triggered if the user asks for a deep dive, or specifies a book part and page. The LLM intelligently navigates the previously defined TOC JSON tree to fetch specific, contiguous pages.

* **Verification and Output Generation:** * The LLM enforces strict citation mechanisms natively structured from the chunk metadata (e.g., ”(Page 584, Vol 1)”).

* It generates its final response in a strictly validated ‘JSON‘ format ensuring three

keys: ‘answer‘ (the theological ruling in markdown), ‘relevant$_{q}$*uestions‘(generating3contextuallyawar*
updiscussionpointstokeeptheuserengagedinalearningloop),and‘resources‘(anarraytrackingallacc
AcontinuousDatabaselayer(‘TocManager‘)quietlyinterceptstheLLM′stoolusages,storingtokenco

Chapter Transition: Having mapped the theoretical limitations of base LLMs and successfully architectured an autonomous Agentic system capable of resolving complex Islamic rulings safely, the focus shifts to quantifying these achievements. The final chapter strictly consolidates the evaluation matrices, performance statistics, and broader developmental results attained over the entire research spanning from the early predictive NLP algorithms to the absolute success of the Agentix RAG system.

Chapter B

Discussion and Synthesized Validation

Chapter Overview: *While the preceding chapters isolated individual empirical experiments—from transition-word NLP extraction to the Fatawa validation matrices—this chapter synthesizes the holistic findings. It discusses the broader validation of the journey, the significance of the developed datasets, and the definitive proof that Agentic RAG architectures successfully bridge the gap between volatile generative AI and dogmatic religious rigor.*

B.1 The Evolution from Extraction to Generation

The foundational research validated that traditional AI was exceptional at structured classification but incapable of native reasoning. The hypothesis that Generative AI could cross this barrier was aggressively tested, revealing a paradoxical result: base LLMs possess vast linguistic reasoning capabilities, yet they fail at strict factual recall in niche domains (averaging only $\sim 46\%$ accuracy against Hanafi domain experts).

The Agentix Islam RAG framework resolved this paradox by completely stripping the LLM of its responsibility to act as a ”knowledge base,” demoting it instead to a highly efficient ”reasoning engine” that operates exclusively over the strictly provided vector context.

B.2 Dataset Contributions and Hallucination Reduction

A core outcome of this research is the quantitative mitigation of hallucination, which was made possible almost entirely through the engineering of high-quality datasets.

1. **The Ahadith Authenticity Dataset:** Separating the Sanad from the Matn and utilizing expert LLM-translation established the first doctrinally clean schema for semantic text processing in Hadith. The validation via expert theologians proved that computational reranking logic could correctly map to human theological priorities.

2. **The Hanafi Fatwa Dataset:** Categorizing 10,000 empirical rulings established a gold-standard baseline previously absent from Arabic NLP computational theology, enabling the strict LLM-as-a-judge benchmarking.

B.3 Validating the RAG Architectural Superiority

By forcing the generative model to append citations directly derived from the metadata (e.g., specific Page and Volume numbers), the architecture enacted a systemic verification mechanism. Because the system was explicitly constrained to output ”Information Not Available” rather than guessing when a definitive node was not retrieved, the structural hallucination rate dropped effectively to zero across the 4,000 operational queries submitted by live users.

Chapter C

Conclusion

This dissertation successfully maps and navigates the profound complexities of applying artificial intelligence to specialized, highly sensitive educational domains. The journey began by establishing traditional NLP baselines and acknowledging the limitations of pre-generative AI through a massive systematic review. As the research transitioned into the zero-tolerance domain of Islamic Jurisprudence, it definitively proved that relying on open-ended generative knowledge bases (like raw GPT or Gemini) is fundamentally insufficient for maintaining scholarly doctrine.

However, by rigorously structuring classical Arabic text, isolating structural components (Matn vs. Sanad), capturing complex TOC hierarchies, and layering Semantic Search over Contextual Reranking, this research engineered a definitive solution. The deployed ”Agentix Islam RAG” framework proves that sophisticated LLMs can be mathematically constrained by structural data pipelines to operate as perfectly reliable, non-hallucinating academic assistants. In doing so, this research has not only modernized accessibility to ancient theological texts but has established a replicable, validated architectural blueprint for deploying Generative AI safely into any highly authoritative domain.

Bibliography

[1] C. S. Rajender Kumar Surana, Shriya, D. B. Gupta, and S. P. Shankar, "Intelligent chatbot for requirements elicitation and classification," in *2019 4th International Conference on Recent Trends on Electronics, Information, Communication & Technology (RTEICT)*, 2019, pp. 866–870. DOI: 10.1109/RTEICT46194.2019.9016907

[2] L. Erlenhov, F. Gomes de Oliveira Neto, R. Scandariato, and P. Leitner, "Current and future bots in software development," in *2019 IEEE/ACM 1st International Workshop on Bots in Software Engineering (BotSE)*, 2019, pp. 7–11. DOI: 10.1109/BotSE.2019.00009

[3] D. S. Battina, "Artificial intelligence in software test automation: A systematic literature review," *International Journal of Emerging Technologies and Innovative Research (www. jetir. org| UGC and issn Approved), ISSN*, pp. 2349–5162, 2019.

[4] Y. Xu, C.-H. Shieh, P. van Esch, and I.-L. Ling, "Ai customer service: Task complexity, problem-solving ability, and usage intention," *Australasian marketing journal*, vol. 28, no. 4, pp. 189–199, 2020.

[5] D. Khurana, A. Koli, K. Khatter, and S. Singh, "Natural language processing: State of the art, current trends and challenges," *Multimedia tools and applications*, vol. 82, no. 3, pp. 3713–3744, 2023.

[6] A. Di Sorbo, C. A. Visaggio, M. Di Penta, G. Canfora, and S. Panichella, "An nlp-based tool for software artifacts analysis," in *2021 IEEE International Conference on Software Maintenance and Evolution (ICSME)*, 2021, pp. 569–573. DOI: 10.1109/ICSME52107.2021.00058

[7] F. N. A. Al Omran and C. Treude, "Choosing an nlp library for analyzing software documentation: A systematic literature review and a series of experiments," in *2017 IEEE/ACM 14th International Conference on Mining Software Repositories (MSR)*, 2017, pp. 187–197. DOI: 10.1109/MSR.2017.42

[8] A. Ferrari and A. Esuli, "An nlp approach for cross-domain ambiguity detection in requirements engineering," *Automated Software Engineering*, vol. 26, pp. 559–598, Sep. 2019. DOI: 10.1007/s10515-019-00261-7

[9] S. Joshi and D. Deshpande, "Textual requirement analysis for uml diagram extraction by using nlp," *International Journal of Computer Applications*, vol. 50, pp. 42–46, Jul. 2012. DOI: 10.5120/7795-0906

[10] C. J. Rameshbhai and J. Paulose, "Opinion mining on newspaper headlines using svm and nlp," *International Journal of Electrical and Computer Engineering (IJECE)*, vol. 9, no. 3, pp. 2152–2163, 2019.

[11] M. R. Hasan, M. Maliha, and M. Arifuzzaman, "Sentiment analysis with nlp on twitter data," in *2019 International Conference on Computer, Communication, Chemical, Materials and Electronic Engineering (IC4ME2)*, 2019, pp. 1–4. DOI: 10.1109/IC4ME247184.2019.9036670

[12] D. Hendrycks, C. Burns, A. Chen, and S. Ball, "Cuad: An expert-annotated nlp dataset for legal contract review," *arXiv preprint arXiv:2103.06268*, 2021.

[13] V. Carchiolo, A. Longheu, G. Reitano, and L. Zagarella, "Medical prescription classification: A nlp-based approach," in *2019 Federated Conference on Computer Science and Information Systems (FedCSIS)*, 2019, pp. 605–609. DOI: 10.15439/2019F197

[14] S. Sarica, B. Song, E. Low, and J. Luo, "Engineering knowledge graph for keyword discovery in patent search," *Proceedings of the Design Society: International Conference on Engineering Design*, vol. 1, no. 1, pp. 2249–2258, 2019. DOI: 10.1017/dsi.2019.231

[15] D. Kauchak, G. Leroy, M. Pei, and S. Colina, "Predicting transition words between sentence for english and spanish medical text," in *AMIA Annual Symposium Proceedings*, American Medical Informatics Association, vol. 2019, 2019, p. 523.

[16] F. Ouyang, L. Zheng, and P. Jiao, "Artificial intelligence in online higher education: A systematic review of empirical research from 2011 to 2020," *Educational Information Technology*, vol. 27, pp. 7893–7925, 2022. DOI: 10.1007/s10639-022-10925-9 [Online]. Available: https://doi.org/10.1007/s10639-022-10925-9

[17] F. Tahiru, "Ai in education: A systematic literature review," *Journal of Cases on Information Technology*, vol. 23, pp. 1–20, Jan. 2021. DOI: 10.4018/JCIT.2021010101

[18] T. K. Chiu, Q. Xia, X. Zhou, C. S. Chai, and M. Cheng, "Systematic literature review on opportunities, challenges, and future research recommendations of artificial intelligence in education," *Computers and Education: Artificial Intelligence*, vol. 4, p. 100 118, 2023, ISSN: 2666-920X. DOI: https://doi.org/10.1016/j.caeai.2022.100118 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2666920X2200073X

[19] M. Farina, A. Gorb, A. Kruglov, and G. Succi, "Technologies for gqm-based metrics recommender systems: A systematic literature review," *IEEE Access*, vol. 10, pp. 23 098–23 111, 2022. DOI: 10.1109/ACCESS.2022.3152397

[20] N. Grönberg, A. Knutas, T. Hynninen, and M. Hujala, “Palaute: An online text mining tool for analyzing written student course feedback,” *IEEE Access*, vol. 9, pp. 134 518–134 529, 2021. DOI: 10.1109/ACCESS.2021.3116425

[21] G. I. Nitin, G. Swapna, and V. Shankararaman, “Analyzing educational comments for topics and sentiments: A text analytics approach,” in *2015 IEEE Frontiers in Education Conference (FIE)*, 2015, pp. 1–9. DOI: 10.1109/FIE.2015.7344296

[22] S. Cunningham-Nelson, M. Baktashmotlagh, and W. Boles, “Visualizing student opinion through text analysis,” *IEEE Transactions on Education*, vol. 62, no. 4, pp. 305–311, 2019. DOI: 10.1109/TE.2019.2924385

[23] X. Yang-cai and W. Rui, “A study of mooc course review topics mining based on lda topic model,” in *Proceedings of the 3rd Africa-Asia Dialogue Network (AADN) International Conference on Advances in Business Management and Electronic Commerce Research*, ser. AADNIC-ABMECR ’21, Ganzhou, China: Association for Computing Machinery, 2022, pp. 44–48, ISBN: 9781450390415. DOI: 10.1145/3503491.3503498 [Online]. Available: https://doi.org/10.1145/3503491.3503498

[24] C. Guo and Y. Liu, “Bp neural network-based evaluation on university teachers’ teaching quality,” in *Proceedings of the 2020 3rd International Conference on E-Business, Information Management and Computer Science*, ser. EBIMCS ’20, Wuhan, China: Association for Computing Machinery, 2021, pp. 285–289, ISBN: 9781450389099. DOI: 10.1145/3453187.3453348 [Online]. Available: https://doi.org/10.1145/3453187.3453348

[25] S. Pyasi, S. Gottipati, and V. Shankararaman, “Sufat - an analytics tool for gaining insights from student feedback comments,” in *2018 IEEE Frontiers in Education Conference (FIE)*, 2018, pp. 1–9. DOI: 10.1109/FIE.2018.8658457

[26] H. E. Abdelkader, A. G. Gad, A. A. Abohany, and S. E. Sorour, “An efficient data mining technique for assessing satisfaction level with online learning for higher

education students during the covid-19," *IEEE Access*, vol. 10, pp. 6286–6303, 2022. DOI: 10.1109/ACCESS.2022.3143035

[27] F. F. Lubis, Y. Rosmansyah, and S. H. Supangkat, "Experience in learners review to determine attribute relation for course completion," in *2016 International Conference on ICT For Smart Society (ICISS)*, 2016, pp. 32–36. DOI: 10.1109/ICTSS.2016.7792865

[28] K. Muollo, P. Basavaraj, and I. Garibay, "Understanding students' online reviews to improve college experience and graduation rates of stem programs at the largest post-secondary institution: A learner-centered study," in *2018 IEEE Frontiers in Education Conference (FIE)*, 2018, pp. 1–7. DOI: 10.1109/FIE.2018.8658450

[29] G. Elia, G. Solazzo, G. Lorenzo, and G. Passiante, "Assessing learners' satisfaction in collaborative online courses through a big data approach," *Computers in Human Behavior*, vol. 92, pp. 589–599, 2019, ISSN: 0747-5632. DOI: https://doi.org/10.1016/j.chb.2018.04.033 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0747563218301973

[30] Z. Nasim, Q. Rajput, and S. Haider, "Sentiment analysis of student feedback using machine learning and lexicon based approaches," in *2017 International Conference on Research and Innovation in Information Systems (ICRIIS)*, 2017, pp. 1–6. DOI: 10.1109/ICRIIS.2017.8002475

[31] K. F. Hew, X. Hu, C. Qiao, and Y. Tang, "What predicts student satisfaction with moocs: A gradient boosting trees supervised machine learning and sentiment analysis approach," *Computers Education*, vol. 145, p. 103 724, 2020, ISSN: 0360-1315. DOI: https://doi.org/10.1016/j.compedu.2019.103724 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0360131519302775

[32] P. Gou and S. Song, "Research on project-based teaching quality evaluation based on fuzzy cluster analysis," in *2019 14th International Conference on Computer*

Science Education (ICCSE), 2019, pp. 826–829. DOI: 10.1109/ICCSE.2019.8845364

[33] J. S. Cross, N. Keerativoranan, M. K. J. Carlon, Y. H. Tan, Z. Rakhimberdina, and H. Mori, "Improving mooc quality using learning analytics and tools," in *2019 IEEE Learning With MOOCS (LWMOOCS)*, 2019, pp. 174–179. DOI: 10.1109/LWMOOCS47620.2019.8939617

[34] V. Heilala, M. Saarela, P. Jääskelä, and T. Kärkkäinen, "Course satisfaction in engineering education through the lens of student agency analytics," in *2020 IEEE Frontiers in Education Conference (FIE)*, 2020, pp. 1–9. DOI: 10.1109/FIE44824.2020.9274141

[35] A. B. Firdausiah Mansur, N. Yusof, and A. H. Basori, "Personalized learning model based on deep learning algorithm for student behaviour analytic," *Procedia Computer Science*, vol. 163, pp. 125–133, 2019, 16th Learning and Technology Conference 2019Artificial Intelligence and Machine Learning: Embedding the Intelligence, ISSN: 1877-0509. DOI: https://doi.org/10.1016/j.procs.2019.12.094 [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1877050919321337

[36] L. Balachandran and A. Kirupananda, "Online reviews evaluation system for higher education institution: An aspect based sentiment analysis tool," in *2017 11th International Conference on Software, Knowledge, Information Management and Applications (SKIMA)*, 2017, pp. 1–7. DOI: 10.1109/SKIMA.2017.8294118

[37] C. Wan, H. Kang, F. Jin, and X. L. Zhu, "Competition neural network model application in the bilingual teaching quality evaluation of universities," in *2020 5th International Conference on Mechanical, Control and Computer Engineering (ICMCCE)*, 2020, pp. 1289–1292. DOI: 10.1109/ICMCCE51767.2020.00282

[38] A. Romanov et al., “Applying ai in education creating a grading prediction system and digitalizing student profiles,” in *2022 International Conference on Frontiers of Communications, Information System and Data Science (CISDS)*, 2022, pp. 84–93. DOI: 10.1109/CISDS57597.2022.00021

[39] J. Wanyin, “Research on the application of bp neural network in vocal music teaching quality evaluation,” in *2020 5th International Conference on Smart Grid and Electrical Automation (ICSGEA)*, 2020, pp. 306–309. DOI: 10.1109/ICSGEA51094.2020.00072

[40] C. Beiqing and Z. Chunrong, “Design of online teaching quality evaluation system for private university: —research based on deep learning algorithm,” in *2020 International Conference on Modern Education and Information Management (ICMEIM)*, 2020, pp. 29–32. DOI: 10.1109/ICMEIM51375.2020.00015

[41] B. Huang and K. Wang, “An improved bp neural network-based quality evaluation model for chinese international education teaching courses,” in *2022 The 6th International Conference on Compute and Data Analysis*, ser. ICCDA 2022, Shanghai, China: Association for Computing Machinery, 2022, pp. 122–127, ISBN: 9781450395472. DOI: 10.1145/3523089.3523109 [Online]. Available: https://doi.org/10.1145/3523089.3523109

[42] F. Zhang, Q. Jiang, C. Jia, and B. Zhou, “Evaluation model of pe teaching quality based on active learning support vector machine,” in *Proceedings of the 5th International Conference on Frontiers of Educational Technologies*, ser. ICFET 2019, Beijing, China: Association for Computing Machinery, 2019, pp. 101–104, ISBN: 9781450362931. DOI: 10.1145/3338188.3338202 [Online]. Available: https://doi.org/10.1145/3338188.3338202

[43] P. Chen, “Evaluation system of tourism psychology teaching quality based on convolutional neural network,” in *2021 4th International Conference on Information Systems and Computer Aided Education*, ser. ICISCAE 2021, Dalian, China:

Association for Computing Machinery, 2021, pp. 964–967, ISBN: 9781450390255. DOI: 10.1145/3482632.3483062 [Online]. Available: https://doi.org/10.1145/3482632.3483062

[44] Z. Ming, "An evaluation model of moral education teaching quality in universities based on system control theory," in *2021 13th International Conference on Measuring Technology and Mechatronics Automation (ICMTMA)*, 2021, pp. 535–538. DOI: 10.1109/ICMTMA52658.2021.00123

[45] X. Shen and T. Bian, "Research on the application of bp neural network in the evaluation of college teaching quality," in *Proceedings of the 2022 6th International Conference on E-Education, E-Business and E-Technology*, ser. ICEBT '22, Beijing, China: Association for Computing Machinery, 2022, pp. 73–77, ISBN: 9781450397216. DOI: 10.1145/3549843.3549854 [Online]. Available: https://doi.org/10.1145/3549843.3549854

[46] H. Mei-zhong, "The evaluation and application of bp neural network in the multiple quality assurance paths of higher education," in *Proceedings of the 2021 4th International Conference on E-Business, Information Management and Computer Science*, ser. EBIMCS '21, Hong Kong, China: Association for Computing Machinery, 2022, pp. 467–474, ISBN: 9781450395687. DOI: 10.1145/3511716.3511786 [Online]. Available: https://doi.org/10.1145/3511716.3511786

[47] M. Imtiyaz and A. Singhal, "Evaluating the quality of teaching in higher education institutes using clustering approach and set pair analysis," in *2015 1st International Conference on Next Generation Computing Technologies (NGCT)*, 2015, pp. 588–592. DOI: 10.1109/NGCT.2015.7375189

[48] J. Zhan, X. Fan, and Y. Zhao, "Automatic rules generation for teaching strategies improvement," in *Proceedings of the 4th International Conference on Computer Science and Application Engineering*, ser. CSAE '20, Sanya, China: As-

sociation for Computing Machinery, 2020, ISBN: 9781450377720. DOI: 10.1145/3424978.3425079 [Online]. Available: https://doi.org/10.1145/3424978.3425079

[49] Z. Niu, W. Li, X. Yan, and N. Wu, “Exploring causes for the dropout on massive open online courses,” in *Proceedings of ACM Turing Celebration Conference - China*, ser. TURC ’18, Shanghai, China: Association for Computing Machinery, 2018, pp. 47–52, ISBN: 9781450364157. DOI: 10.1145/3210713.3210727 [Online]. Available: https://doi.org/10.1145/3210713.3210727

[50] Z. Zhan, T. Shen, L. Jin, F. Huang, and H. Xu, “Research on evaluation of online teaching effect based on deep learning technology,” in *2021 IEEE 5th Advanced Information Technology, Electronic and Automation Control Conference (IAEAC)*, 2021, pp. 249–253. DOI: 10.1109/IAEAC50856.2021.9390652

[51] C. Durach, J. Kembro, and A. Wieland, “A new paradigm for systematic literature reviews in supply chain management,” *Journal of Supply Chain Management*, vol. 53, Oct. 2017. DOI: 10.1111/jscm.12145

[52] B. Kitchenham, “Procedures for performing systematic reviews,” *Keele, UK, Keele Univ.*, vol. 33, Aug. 2004.

[53] M. Yusram, “Al-manẓūmah al-bayqūniyyah wa-ʿināyat ahl al-ʿilm bihā,” *Al-Baṣīrah: Majallat al-Dirāsāt al-Islāmiyyah*, vol. 2, no. 1, pp. 127–167, 2021. DOI: 10.36701/bashirah.v2i1.333 [Online]. Available: https://journal.stiba.ac.id/index.php/bashirah/article/view/333

[54] M. Mazzara et al., “Education after covid-19,” in *Smart and Sustainable Technology for Resilient Cities and Communities*, R. J. Howlett, L. C. Jain, J. R. Littlewood, and M. M. Balas, Eds. Singapore: Springer Nature Singapore, 2022, pp. 193–207, ISBN: 978-981-16-9101-0. DOI: 10.1007/978-981-16-9101-0_14 [Online]. Available: https://doi.org/10.1007/978-981-16-9101-0_14

[55] A. Romanov et al., “Applying ai in education creating a grading prediction system and digitalizing student profiles,” in *2022 International Conference on Frontiers of Communications, Information System and Data Science (CISDS)*, 2022, pp. 84–93. DOI: 10.1109/CISDS57597.2022.00021

[56] S. H. Altammami, “Artificial intelligence for understanding the hadith,” Ph.D. dissertation, University of Leeds, 2023. [Online]. Available: https://etheses.whiterose.ac.uk/id/eprint/32802

[57] M. D. M. Tahir Ismail and M. A. Faisal, *Imam al-nawawi and his approach in his book (riyad al-salihin)*. [Online]. Available: https://jcois.uobaghdad.edu.iq/index.php/2075_8626/article/view/251

[58] S. Hasan, *An Introduction to the Science of Hadith*. Al-Quran Society, 1994.

[59] K. Bilal and S. Mohsin, “Muhadith: A cloud based distributed expert system for classification of ahadith,” in *2012 10th International Conference on Frontiers of Information Technology*, IEEE, 2012, pp. 73–78.

[60] S. Hakak et al., “Digital hadith authentication: Recent advances, open challenges, and future directions,” *Transactions on Emerging Telecommunications Technologies*, vol. 33, p. 3977, 2022.

[61] K. Aldhaln, A. Zeki, A. Zeki, and H. Alreshidi, “Improving knowledge extraction of hadith classifier using decision-tree algorithm,” in *International Conference on Information Retrieval & Knowledge Management*, IEEE, 2012, pp. 148–152.

[62] Y. ibn Sharaf An-Nawawi, *Riyad as-Salihin*. n.d. Translated by [Translator’s Name], Publisher: [Publisher’s Name], [Year of Publication].

[63] H. Bast, B. Buchhold, and E. Haussmann, “Semantic search on text and knowledge bases,” *Foundations and Trends® in Information Retrieval*, vol. 10, no. 2-3, pp. 119–271, 2016. DOI: 10.1561/1500000032 [Online]. Available: http://dx.doi.org/10.1561/1500000032

[64] H. Mubarak, R. Malhas, W. Mansour, et al., "Islamiceval 2025: The first shared task of capturing llms hallucination in islamic content," in *Proceedings of ArabicNLP 2025*, ACL Anthology, 2025.

[65] E. Elmahjub, J. Qadir, A. Mushtaq, R. Naeem, I. Ghaznavi, and W. Iqbal, "Islamiclegalbench: Evaluating llms knowledge and reasoning of islamic law across 1,200 years of islamic pluralist legal traditions," *arXiv preprint arXiv:2602.21226*, 2026.

[66] M. Atif et al., "Fiqhqa: A benchmark for evaluating llms on islamic jurisprudence," *ResearchGate preprints*, 2025.

[67] Whitethread Institute, *The danger of relying on ai for fatwa writing: A caution for scholars and muftis*, https://whitethread.org/, 2025.